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NOVEMBER 2026

The 2026 Pennsylvania Talent Profile

Emerson Johnston
Amy Zegart

The bright caps, **54 in every 100**, mark the international share of Pennsylvania's STEM master's degrees.



The 2026 Pennsylvania Talent Profile

By Emerson Johnston and Amy Zegart

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Data come from the Integrated Postsecondary Education Data System (IPEDS), the Department of Homeland Security's SEVIS by the Numbers, the National Assessment of Educational Progress, NSF's Survey of Graduate Students and Postdoctorates and Survey of Earned Doctorates, the Department of Labor's Office of Foreign Labor Certification, the USCIS H-1B Employer Data Hub, the Census Bureau's American Community Survey and USPTO patent data via PatentsView, processed in Python; every figure and table is generated from those sources, and each one names its source.

About This Profile

This profile is a companion to *The 2026 US Talent Report: Benchmarking America's STEM Pipeline Amid the Great Talent Competition*. It applies the report's measures to Pennsylvania: the students the state develops at home, those it draws from abroad, and how both compare with the nation and with the states closest to it in size.

Degrees, enrollment, prices, outcomes and faculty come from IPEDS, the federal census of U.S. colleges and universities, harmonized across 1980–2025. Countries of citizenship come from SEVIS by the Numbers, the Department of Homeland Security's monthly count of active F-1 and M-1 students. Each figure names its source. Peer states are the four closest in total fall enrollment.

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Public Data and Tools

The data behind every chart come from federal sources: the Integrated Postsecondary Education Data System (IPEDS), published by the National Center for Education Statistics, as the backbone, together with NAEP, NSF's surveys of graduate students, doctorates and research funding, USAspending.gov, DHS and ICE SEVIS data, the Department of Labor, USCIS, the Census Bureau's American Community Survey and Post-Secondary Employment Outcomes, and USPTO PatentsView. Each figure names its source, and the Sources page lists all of them. An interactive version of this profile accompanies the report.

To share any charts and figures, proper attribution is required. This profile is licensed under Attribution-NoDerivatives 4.0 International (CC BY-ND 4.0). Please cite as: The 2026 Pennsylvania Talent Profile.

AT A GLANCE

What Pennsylvania's numbers say

1. **International students earn 54% of Pennsylvania's STEM master's degrees,** 1.4 points above the U.S. (52%) in 2025, ranking 14th of 51 on dependence. The share is 45% for STEM research doctorates (U.S. 43%) and 8.3% for STEM bachelor's (U.S. 6.6%).
2. **Pennsylvania awarded 5.51K STEM master's degrees to international students in 2025,** 4.7% of the U.S. total and 7th of 51 among states. Since 2015 the count changed +78%, against 2.2× nationally.
3. **The heaviest dependence is in engineering technology,** where international students earned 68% of master's degrees (336 degrees; U.S. 54%). 5 STEM fields with at least 50 master's graduates are majority-international.
4. **Among states of similar size,** Pennsylvania ranks 2nd of 5 on the international share of STEM master's: Pennsylvania 54%, Ohio 58%, Georgia 44%, Arizona 41%, North Carolina 49%.
5. **68% of Pennsylvania's first-time students in fall 2024 were state residents** and 3.4% came from abroad. More students arrived from other states (29.9K) than residents left (22.3K), a net gain of 7.58K; the top destination for those leaving was New York.

54%

of the STEM master's degrees Pennsylvania awards go to international students (U.S. 52%).

SCORECARD

Nine numbers against the nation

637K #7
Students enrolled, 2024
U.S. **19.7M** · −16% since 2015

5.51K #7
International STEM master's, 2025
U.S. **117K** · +78% since 2015

54% #14
International share of STEM master's, 2025
U.S. **52%** · +6.9 pp since 2015

18% #17
International share of grad students, 2024
U.S. **17%** · +2.1 pp since 2015

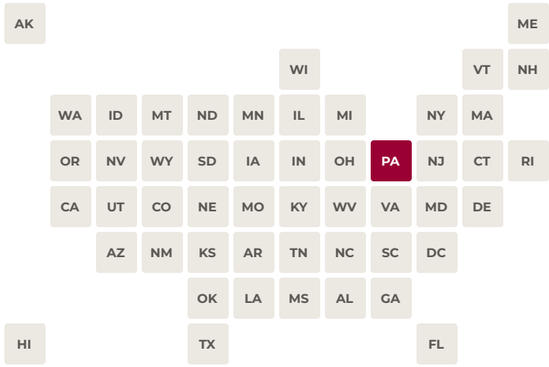
27% #10
STEM share of bachelor's, 2025
U.S. **24%** · +6.2 pp since 2015

68% #15
Six-year graduation rate, 2024
U.S. **65%** · +0.9 pp since 2015

$7.14K #46
Appropriations per student, 2024
U.S. **$11.4K** · +20% since 2015

68% #31
First-time students from in state, 2024
U.S. **–** · −2.4 pp since 2016

$109K #14
Average faculty salary, 2024
U.S. **$106K** · −8.2% since 2016



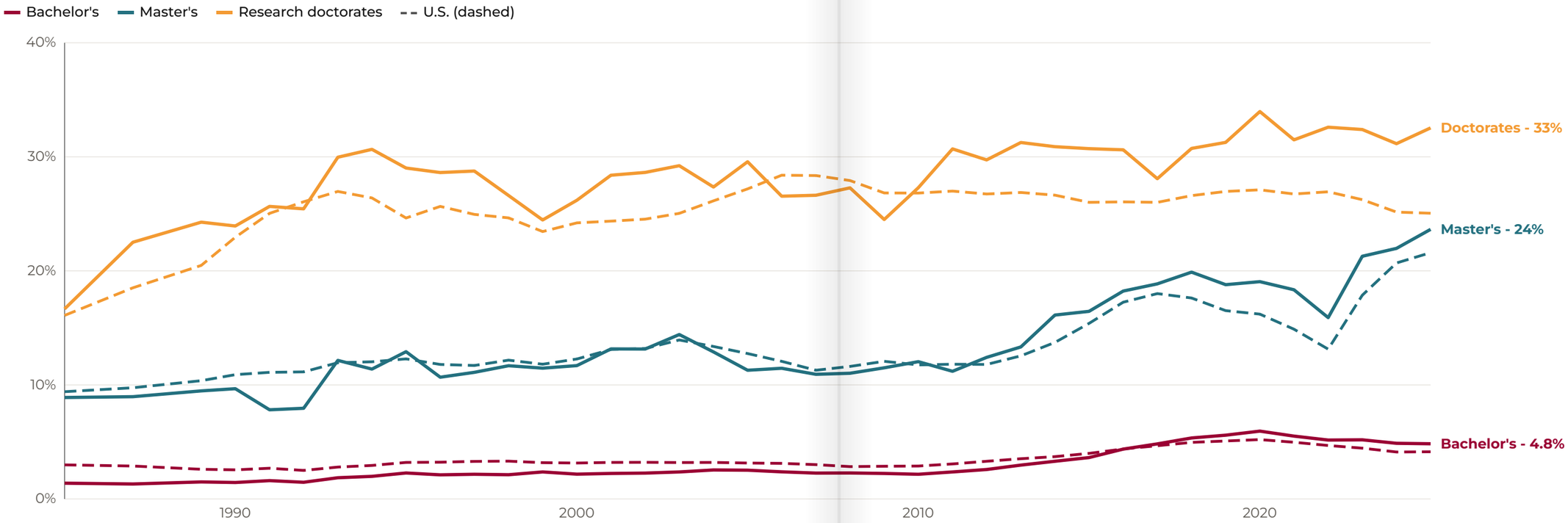
Badges give Pennsylvania's rank among the 50 states and DC (1 = highest). Changes are since 2015.

SECTION 1

International Dependence

Pennsylvania ranks 12th of 51 on the international share of all master's, at 24% in 2025 against 22% for the U.S.

Figure 1 What Share of Pennsylvania's Degrees Go to International Students?



International share of degrees awarded, all fields; state solid, U.S. dashed. Data from NCES, IPEDS Completions, 1985–2025.

International students make up 18% of Pennsylvania's graduate enrollment and 3.9% of its undergraduates. Dependence climbs with each degree: 4.8% of bachelor's, 24% of master's and 33% of research doctorates go to international students.

In STEM the shares are higher still: 8.3% of STEM bachelor's, 54% of STEM master's and 45% of STEM doctorates. The state awards 4.7% of all U.S. international STEM master's, 7th of 51.

18%

Int. share of grad students, 2024

U.S. **17%** · **17th** of 51 · +2.1 pp since 2015

24%

Int. share of all master's, 2025

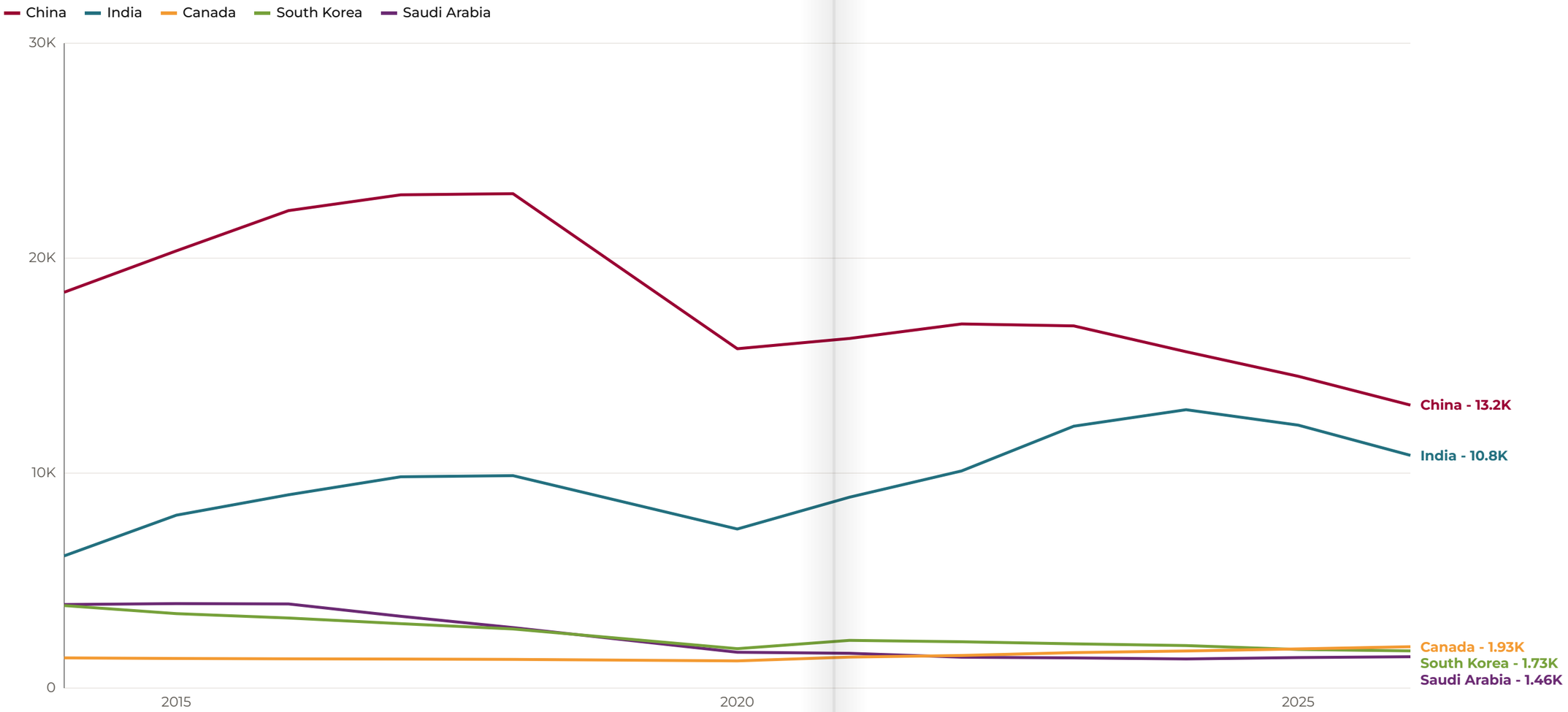
U.S. **22%** · **12th** of 51 · +7.2 pp since 2015

54%

Int. share of STEM master's, 2025

U.S. **52%** · **14th** of 51 · +6.9 pp since 2015

Figure 2 Where Do Pennsylvania's International Students Come From?



Active F-1 and M-1 students by country of citizenship, the five largest in October 2026; one fall snapshot per year (the month closest to November), October 2014–October 2026. All education levels. Data from DHS, SEVIS by the Numbers.

In October 2026, Pennsylvania had 48.6K active international students in SEVIS, 7th of 51. China accounted for 27% of them and India for 22%, against 18% and 27% nationally.

Students came from 186 countries, but the two largest supplied 49%. The state's total is down 7% since 2015, when it stood at 52.4K.

48.6K
Active international students, October 2026
7th of 51 · U.S. **1.19M**

27%
From China
U.S. **18%**

64%
In STEM fields
U.S. **63%**

SECTION 1 · INTERNATIONAL DEPENDENCE

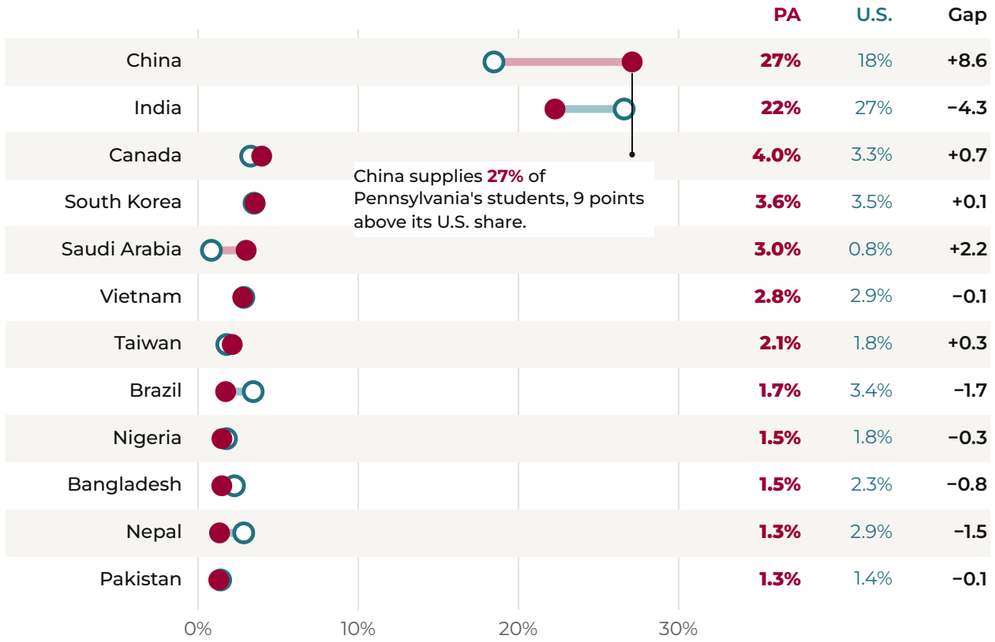
How Does Pennsylvania's Mix of Countries Compare With the Nation's?

In October 2026, Pennsylvania had **48.6K** active F-1 and M-1 students, 7th among the 50 states and DC and **4.1%** of the U.S. total. They came from 186 countries.

China supplied **27%** of them, against 18% nationally, and India **22%**, against 27%. Canada (4.0%), South Korea (3.6%), and Saudi Arabia (3.0%) followed.

Together the two largest countries supplied **49%** of the state's international students (U.S. 45%), 15th among the states, against 54% in 2015.

Figure 3 Is Pennsylvania's Mix of Countries Like the Nation's?



Share of active F-1 and M-1 students, all levels. Data from DHS, SEVIS by the Numbers, October 2026.

48.6K
Active international students, October 2026
7th of 51 · U.S. **1.19M**

27%
From China
U.S. **18%**

22%
From India
U.S. **27%**

186
Countries represented
U.S. **219**

The state's count is down 6.6% from 2025 and down 7.3% since 2015, when it stood at 52.4K. Its share of all U.S. international students is down 0.3 points since 2015, to **4.1%**.

Since 2015, the number of students from China went from 20.3K to 13.2K, India's from 8.04K to 10.8K, and Canada's from 1.38K to 1.93K.

Over the same years, the number from South Korea went from 3.46K to 1.73K and Saudi Arabia's from 3.93K to 1.46K. The table gives each country's count for the first, previous and latest years.

Table 1 International Students in Pennsylvania, 2015–2026

	2015	2025	2026	Change since 2015	U.S. 2026	Rank
ALL INTERNATIONAL STUDENTS						
Active F-1 and M-1 students	52.4K	52K	**48.6K**	−7.3%	1.19M	7th
Share of U.S. active F-1/M-1 students	4.4%	4.1%	**4.1%**	−0.3 pp	–	7th
Countries of citizenship represented	185	185	**186**	+0.5%	219	6th
Share from the two largest countries of origin	54%	51%	**49%**	−4.8 pp	45%	15th
LARGEST COUNTRIES OF CITIZENSHIP (STUDENTS)					**U.S. SHARE**	**STATE SHARE**
China	20.3K	14.5K	**13.2K**	−35%	18%	27%
India	8.04K	12.2K	**10.8K**	+35%	27%	22%
Canada	1.38K	1.82K	**1.93K**	+40%	3.3%	4.0%
South Korea	3.46K	1.79K	**1.73K**	−50%	3.5%	3.6%
Saudi Arabia	3.93K	1.42K	**1.46K**	−63%	0.8%	3.0%

Active F-1 and M-1 students, one fall snapshot per year (the month closest to November; 2019 has none; the latest year is October 2026). Rank among the 50 states and DC. Data from DHS, SEVIS by the Numbers.

Figure 4

Which Countries Supply Pennsylvania's International STEM Students?

31.1K of Pennsylvania's international students were in STEM fields in October 2026, **64%** of the total against 63% nationally, up 27% since 2015.

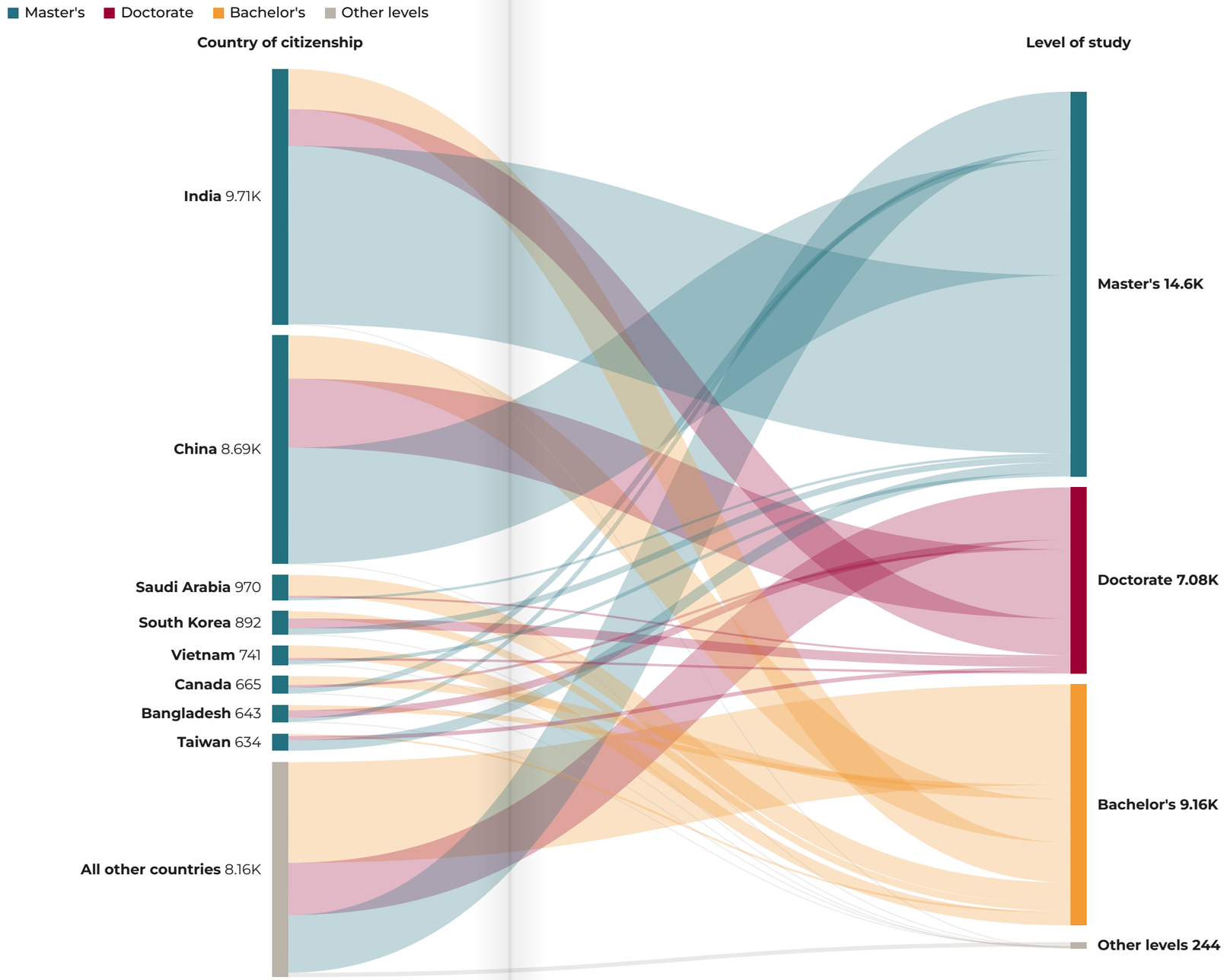
Master's students were the largest group, **14.6K**. India supplied 46% of them and China 30%.

STEM doctoral students went from 4.99K in 2015 to **7.08K**, and STEM master's students from 10.8K to 14.6K.

31.1K
Active STEM F-1/M-1 students, 2026
U.S. **749K** · **7th** of 51 · +27% since 2015

64%
Share in STEM fields
U.S. **63%**

14.6K
Active STEM master's students, 2026
U.S. **393K** · **7th** of 51 · +36% since 2015



Active F-1 and M-1 students in STEM fields by country of citizenship and level of study, October 2026; the eight largest countries are shown. Data from DHS, SEVIS by the Numbers, October 2026.

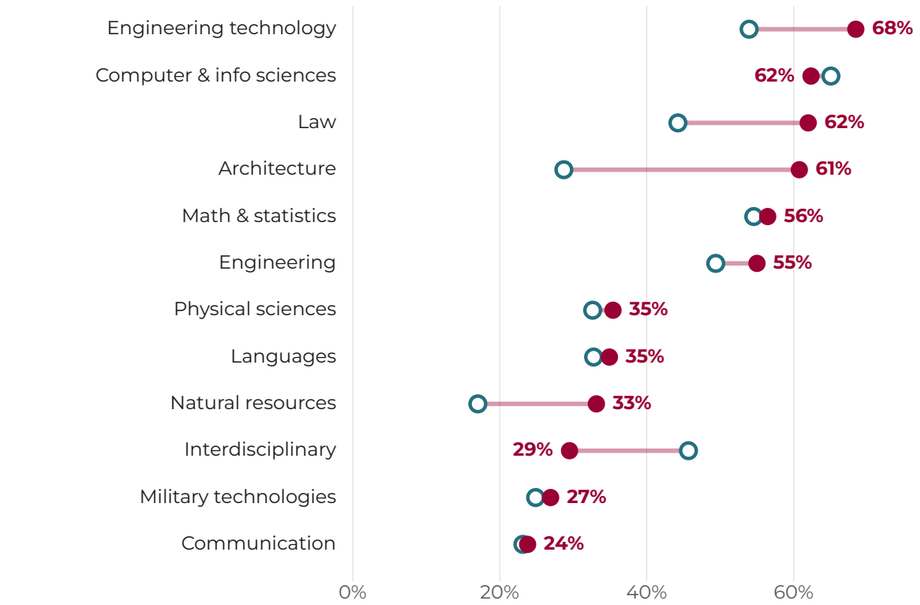
SECTION 1 · INTERNATIONAL DEPENDENCE

How International Are Pennsylvania's Master's Fields?

Each row is a master's field with at least 30 graduates in Pennsylvania in 2025, ranked by the share of degrees earned by international students (U.S. nonresidents on temporary visas). The filled dot is the state's share; the open dot is the national share for the same field.

Highest international shares. International students earned the largest shares of master's degrees in engineering technology (68%, against 54% for the U.S.), computer & info sciences (62%, against 65% for the U.S.), and law (62%, against 44% for the U.S.).

Figure 5 Which Master's Fields Depend Most on International Students?



International share of master's degrees by field, 2025; fields with 30+ graduates. Data from NCES, IPEDS Completions.

24%
Int. share of all master's, 2025
U.S. **22%** · **12th** of 51

54%
Int. share of STEM master's, 2025
U.S. **52%** · **14th** of 51

62%
Int. share of CS master's, 2025
U.S. **65%** · **27th** of 47

The table gives the international share of degrees in each STEM field at three levels: Pennsylvania first, the United States after the slash. The master's column is highlighted. A dash means the state awarded no degrees in the field at that level.

Highest international share among STEM master's fields. International students earned 68% of the master's degrees in engineering technology, against 54% for the U.S.

Fields with an international majority. International students earned more than half of the master's degrees in 5 STEM fields.

Table 2 How International Is Each STEM Field?

Field	Bachelor's	Master's	Doctorates
Computer & info sciences	12% / 8.6%	**62% / 65%**	65% / 56%
Engineering	11% / 7.2%	**55% / 49%**	54% / 55%
Engineering technology	1.6% / 4.5%	**68% / 54%**	– / 42%
Math & statistics	18% / 17%	**56% / 55%**	50% / 53%
Physical sciences	5.4% / 6.0%	**35% / 33%**	42% / 38%
Biological sciences	3.0% / 3.4%	**19% / 23%**	21% / 23%
Natural resources	2.1% / 2.0%	**33% / 17%**	– / 27%
Architecture	7.1% / 7.3%	**61% / 29%**	– / 55%

International share in Pennsylvania / U.S., 2025. Data from NCES, IPEDS Completions.

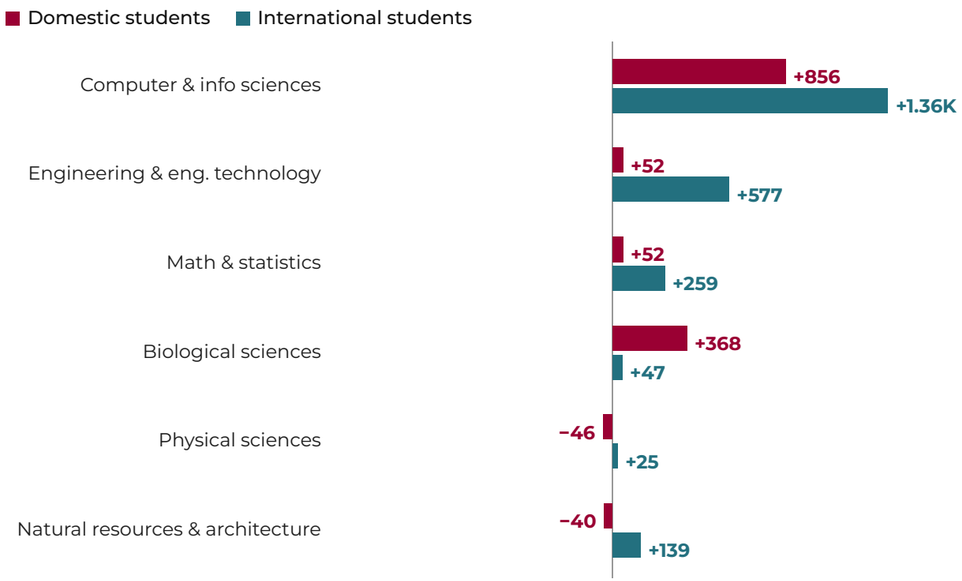
SECTION 1 · INTERNATIONAL DEPENDENCE

How Has Pennsylvania's STEM Master's Output Grown?

Each pair of bars is the change in the number of STEM master's degrees awarded in Pennsylvania between 2015 and 2025 in one field: to domestic students (U.S. citizens and permanent residents) and to international students. Bars to the left of the axis are declines.

STEM master's degrees. The state awarded 10.3K in 2025 and 6.64K in 2015, a change of +2.41K for international students and +1.24K for domestic students.

Figure 6 Where Did the STEM Master's Growth Come From?



Change in STEM master's degrees by field, 2015–2025. Data from NCES, IPEDS Completions.

5.51K
International STEM master's, 2025
U.S. **117K** · **7th** of 51

25%
STEM share of master's, 2025
U.S. **23%** · **15th** of 51

4.7%
Share of U.S. international STEM master's, 2025
7th of 51

5.4
STEM master's per STEM research doctorate, 2025
U.S. **6.4** · **25th** of 51

The table gives the number of STEM master's degrees awarded in Pennsylvania in each field in 2015 and 2025, the change between the two years, the number awarded to international students in 2025, and international students' share of the field's degrees in each year.

STEM share of master's degrees. The share was 25% in 2025, against 23% for the U.S., 15th among the states, up 1.8 points from 23% in 2024 and up 7.0 points since 2015, when it was 18%.

Table 3 How Many STEM Master's Degrees Does Each Field Award?

Field	2015	2025	Change	International 2025	Int. share 2015	Int. share 2025
Computer & info sciences	1.83K	**4.05K**	+121%	2.52K	63%	62%
Engineering & eng. technology	2.94K	**3.57K**	+21%	2.01K	49%	56%
Math & statistics	355	**666**	+88%	376	33%	56%
Biological sciences	767	**1.18K**	+54%	225	23%	19%
Physical sciences	326	**305**	−6.4%	108	25%	35%
Natural resources & architecture	426	**525**	+23%	269	31%	51%

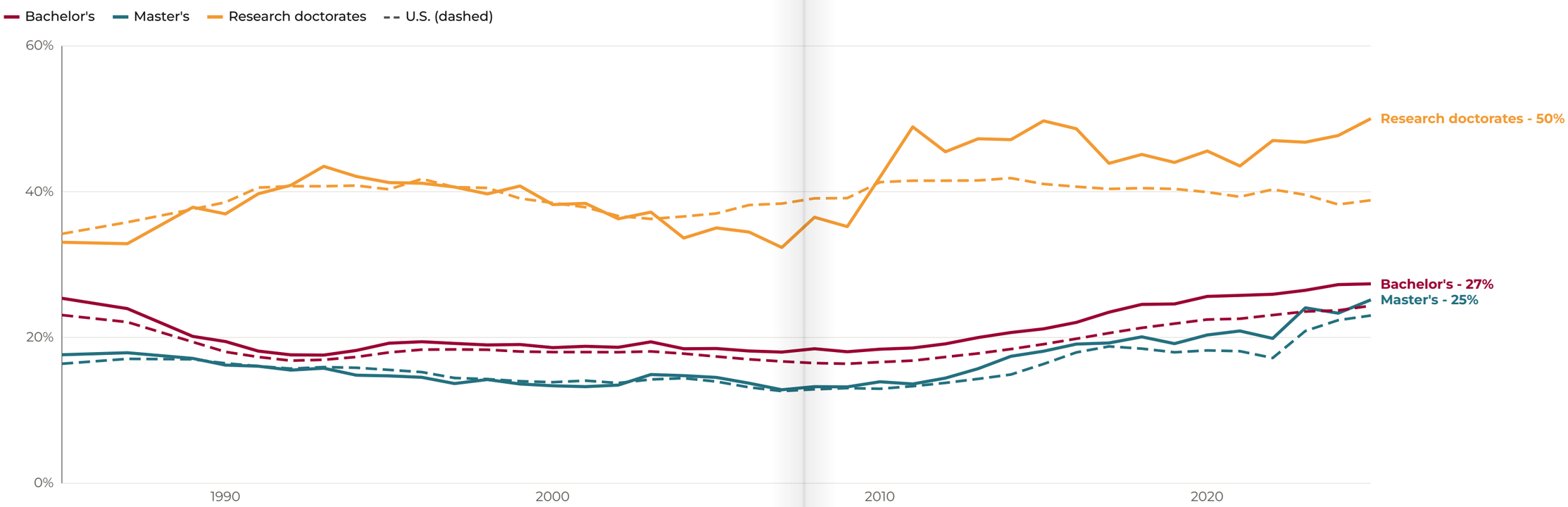
STEM master's degrees awarded in Pennsylvania, all students and international students, 2015 and 2025. Data from NCES, IPEDS Completions.

SECTION 2

STEM Degrees

Pennsylvania ranks 10th of 51 on the STEM share of bachelor's, at 27% in 2025 against 24% for the U.S.

Figure 7 How Much of Pennsylvania's Degree Output Is STEM?



STEM share of degrees by level; state solid, U.S. dashed. Data from NCES, IPEDS Completions.

STEM fields account for 27% of Pennsylvania's bachelor's degrees, 25% of its master's and 50% of its research doctorates (U.S. 24%, 23% and 39%).

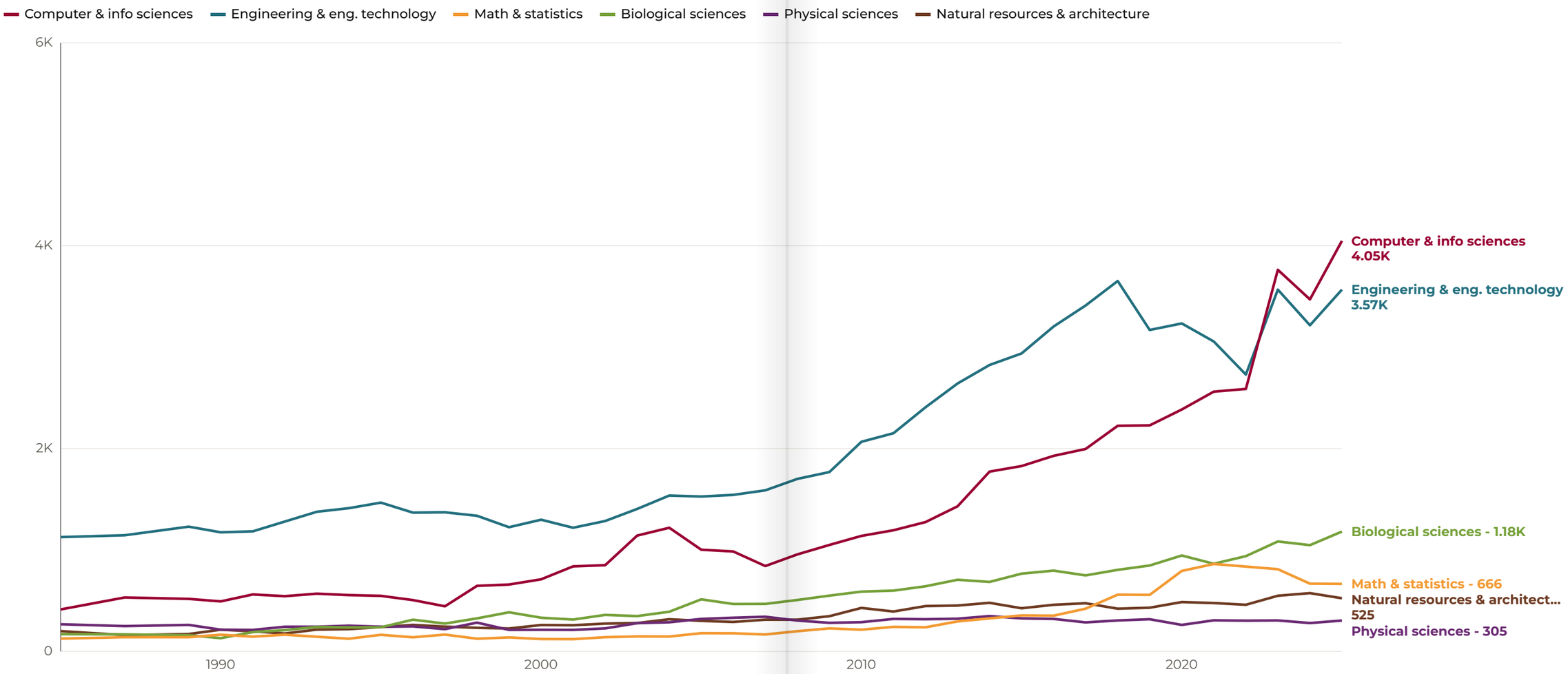
The STEM share of master's degrees is up 7.0 points since 2015, and the state ranks 15th of 51 on it. On the STEM share of bachelor's degrees it ranks 10th of 51.

27%
STEM share of bachelor's, 2025
U.S. **24%** · **10th** of 51 · +6.2 pp since 2015

25%
STEM share of master's, 2025
U.S. **23%** · **15th** of 51 · +7.0 pp since 2015

50%
STEM share of research doctorates, 2025
U.S. **39%** · **10th** of 51 · +0.3 pp since 2015

Figure 8 Which STEM Fields Produce Pennsylvania's Master's Degrees?



STEM master's degrees by field, all students. Data from NCES, IPEDS Completions.

Computer & info sciences produced the most STEM master's degrees in 2025, 4.05K, followed by engineering & eng. technology (3.57K) and biological sciences (1.18K). Master's degrees in computer & info sciences are up 121% since 2015.

International students earned 5.51K of the state's STEM master's degrees in 2025, 4.7% of the national total (7th of 51).

5.51K

International STEM master's, 2025

U.S. **117K** · **7th** of 51 · +78% since 2015

4.7%

Share of U.S. international STEM master's, 2025

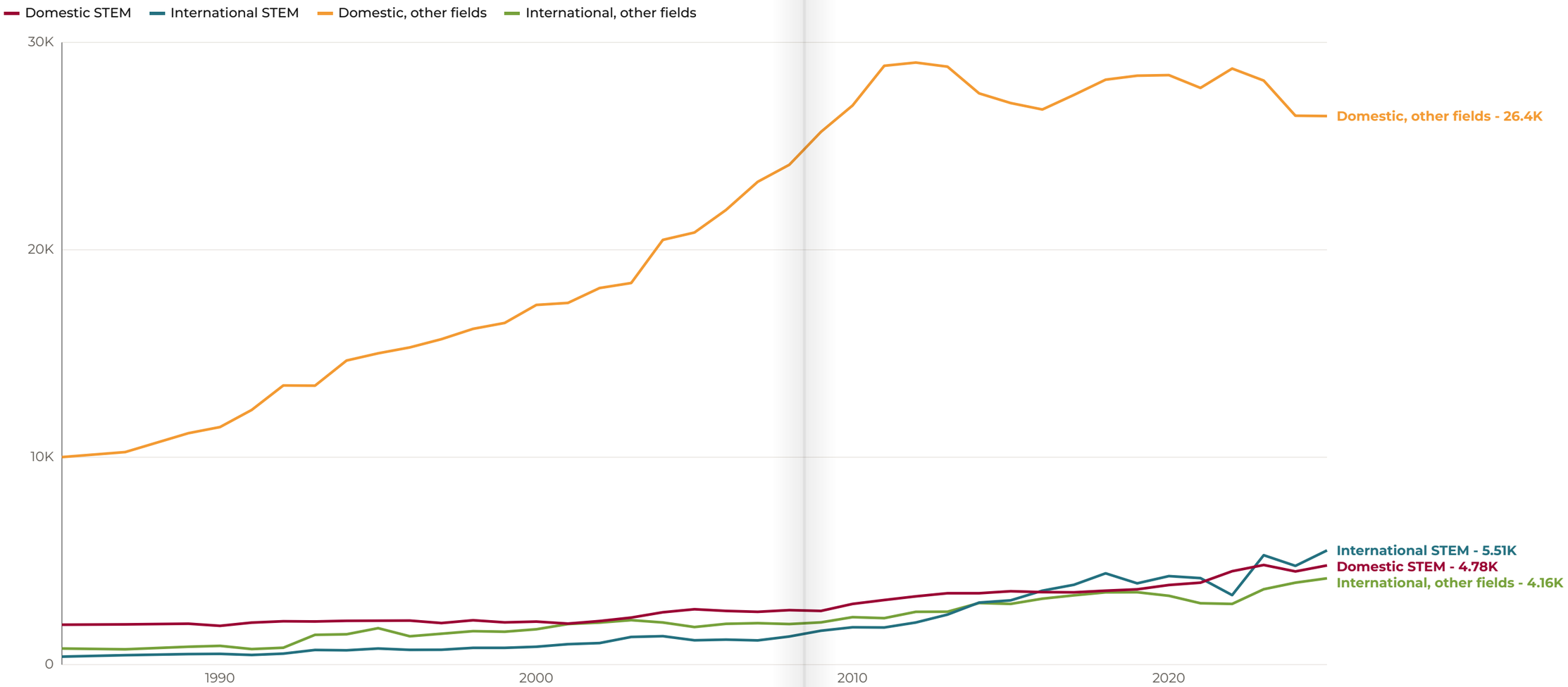
· **7th** of 51 · −1.1 pp since 2015

62%

Int. share of CS master's, 2025

U.S. **65%** · **27th** of 47 · −1.1 pp since 2015

Figure 9 Who Earns Pennsylvania's Master's Degrees?



Master's degrees by student residency and field group. Data from NCES, IPEDS Completions.

Between 2015 and 2025, 2.41K more STEM master's degrees went to international students and 1.24K more to domestic students.

International students earn 54% of Pennsylvania's STEM master's but 14% of master's in other fields (U.S. 52% and 12%).

24%

Int. share of all master's, 2025

U.S. **22%** · **12th** of 51 · +7.2 pp since 2015

54%

Int. share of STEM master's, 2025

U.S. **52%** · **14th** of 51 · +6.9 pp since 2015

5.4

STEM master's per STEM research doctorate (IPEDS), 2025

U.S. **6.4** · **25th** of 51 · +1.2 since 2015

SECTION 3

Where Students Come From

Pennsylvania ranks 16th of 51 on the share of first-time students whose home is outside the US, at 3.4% in 2024 against 3.1% for the U.S.

Figure 10 First-Time Students by Home and Destination, Fall 2024



Home of first-time degree-seeking undergraduates (left) and the state of the institution entered (right); the five largest sources and destinations, the rest grouped. Data from NCES, IPEDS Fall Enrollment.

73.5K residents stayed in Pennsylvania for college in fall 2024; 29.9K students arrived from other states and 3.67K from abroad, while 22.3K residents left.

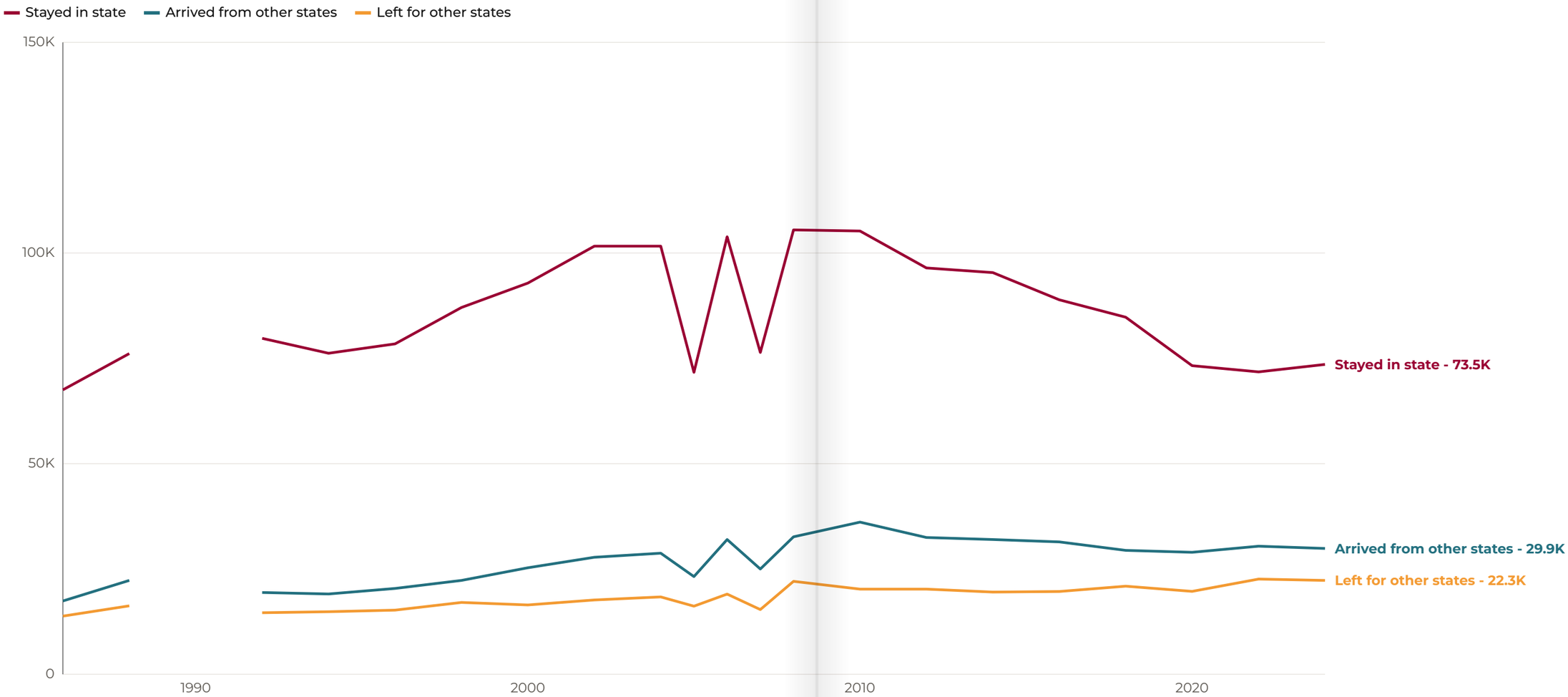
Each band runs from a student's home on the left to the state where they enrolled on the right. 68% of the state's first-time students are residents (U.S. –).

68%
First-time students from in state, 2024
· **31st** of 51 · −2.4 pp since 2016

3.4%
First-time students from abroad, 2024
U.S. **3.1%** · **16th** of 51 · −0.1 pp since 2016

+8
Net migration per 100 residents, 2024
· **21st** of 51 · −2.9 since 2016

Figure 11 How Have the Flows Changed?



First-time degree-seeking undergraduates by flow; even years before 2001. Data from NCES, IPEDS Fall Enrollment.

Since 2015, the number of residents staying in state for college is down 17%, arrivals from other states are down 5%, and residents leaving are up 13%.

Net migration was +8 per 100 resident first-time students (21st of 51). The top destination for residents who left was New York (2.57K).

Figure 12

Where Do Pennsylvania's Newcomers Come From?

The bands show where first-time degree-seeking undergraduates who enrolled at Pennsylvania's institutions in fall 2024 but did not live in the state came from: the 12 largest home states and students from abroad, with all other states grouped. Band widths are proportional to the number of students. Home is the state where the student lived when admitted.

33.5K

First-time students from other states and abroad, fall 2024

Abroad **3.67K**

3.4%

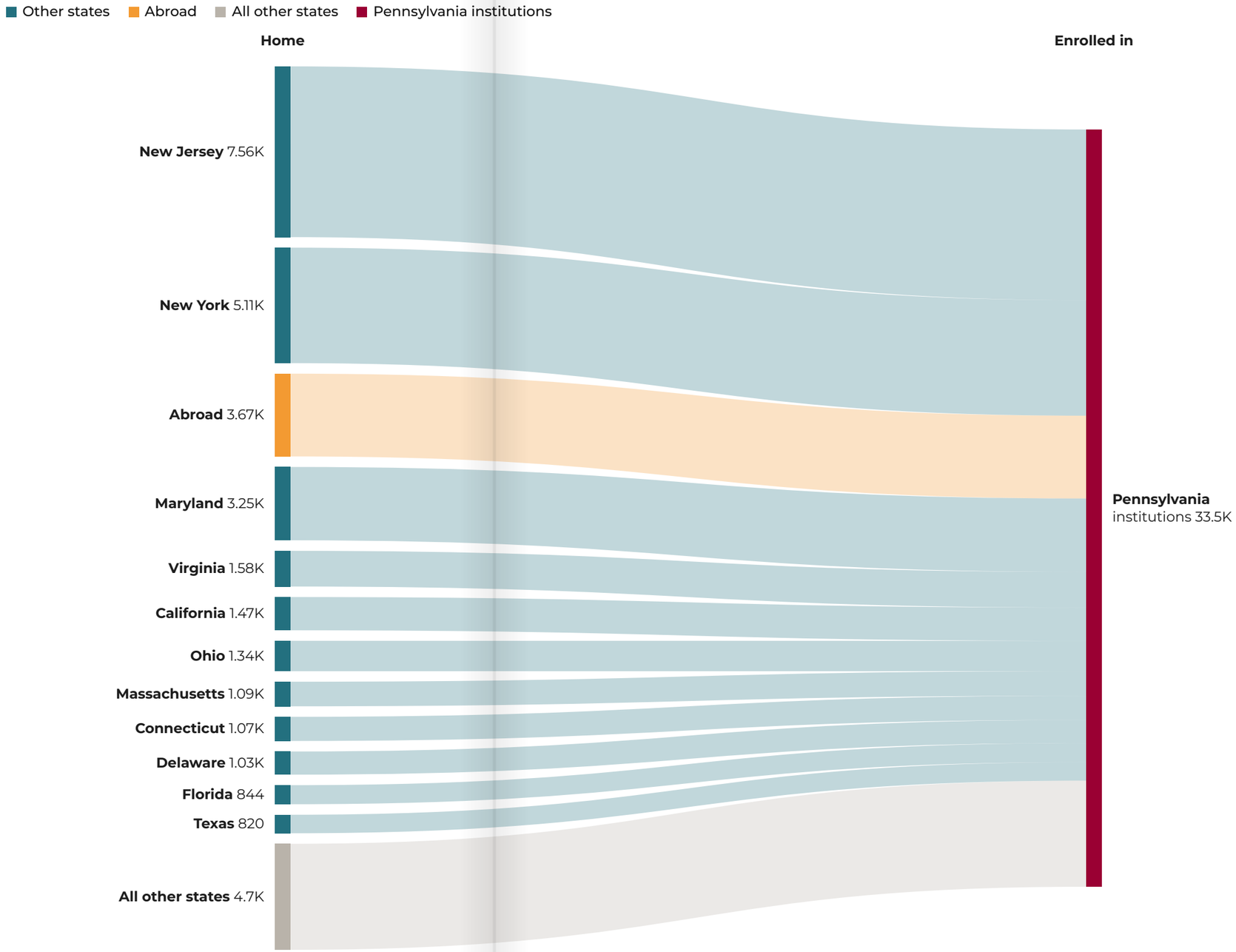
First-time students from abroad, 2024

U.S. **3.1%** · **16th** of 51 · −0.1 pp since 2016

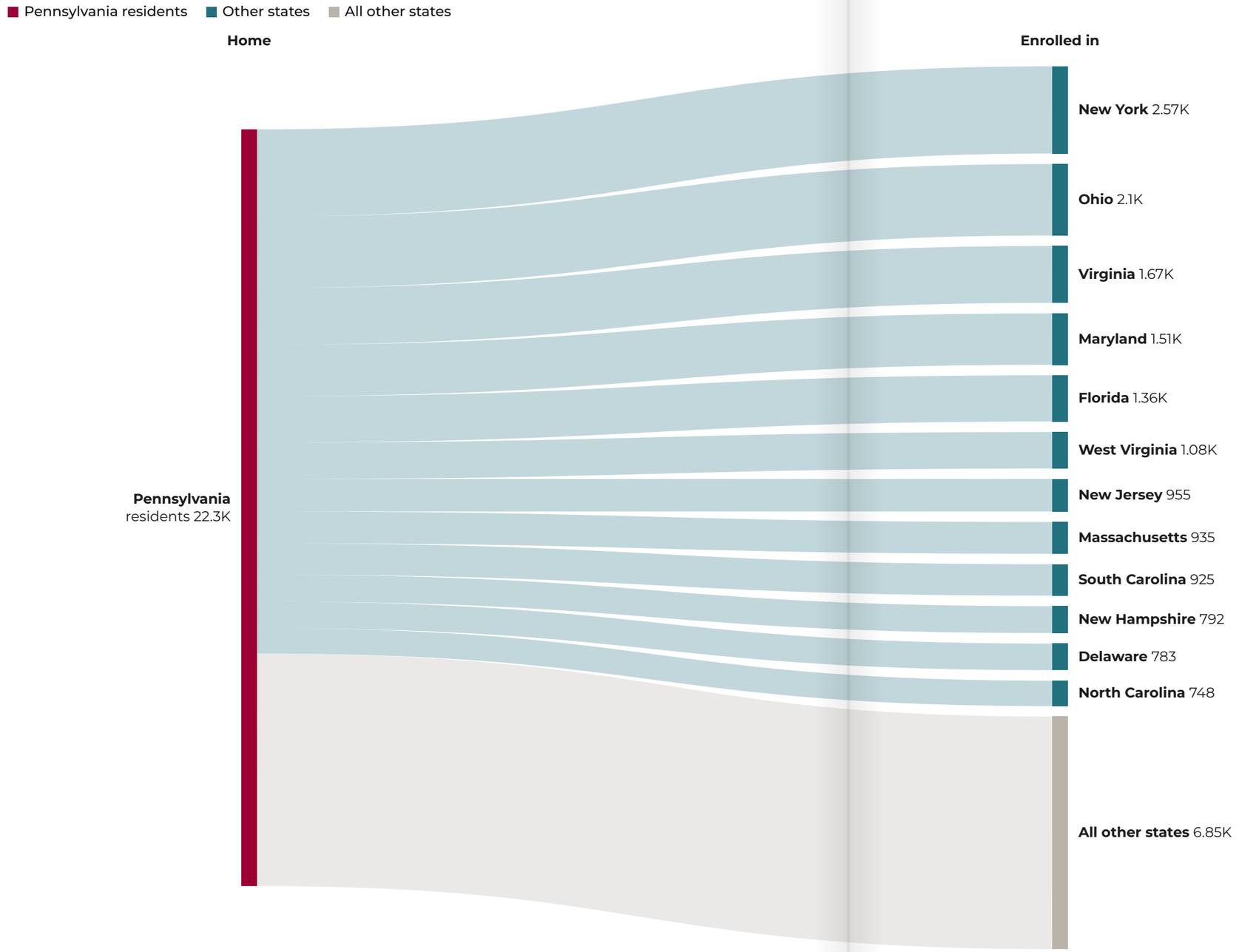
68%

First-time students from in state, 2024

31st of 51 · −2.4 pp since 2016



First-time degree-seeking undergraduates, fall 2024; the 12 largest flows. Data from NCES, IPEDS Fall Enrollment.



First-time degree-seeking undergraduates, fall 2024; the 12 largest flows. Data from NCES, IPEDS Fall Enrollment.

Figure 13

Where Do Pennsylvania's Residents Go When They Leave?

The bands show where Pennsylvania residents who left the state to start college in fall 2024 enrolled: the 12 largest destination states, with all other states grouped. Band widths are proportional to the number of students. Residents who enrolled in the state are not shown.

Residents leaving for other states. The number was 22.3K in fall 2024, down 1.5% from 22.6K in fall 2022 and up 13% since fall 2016, when it was 19.6K.

22.3K
Residents who started college in another state, fall 2024
Arrived from other states **29.9K** · **33.5K** with abroad

+8
Net migration per 100 residents, 2024
21st of 51 · −2.9 since 2016

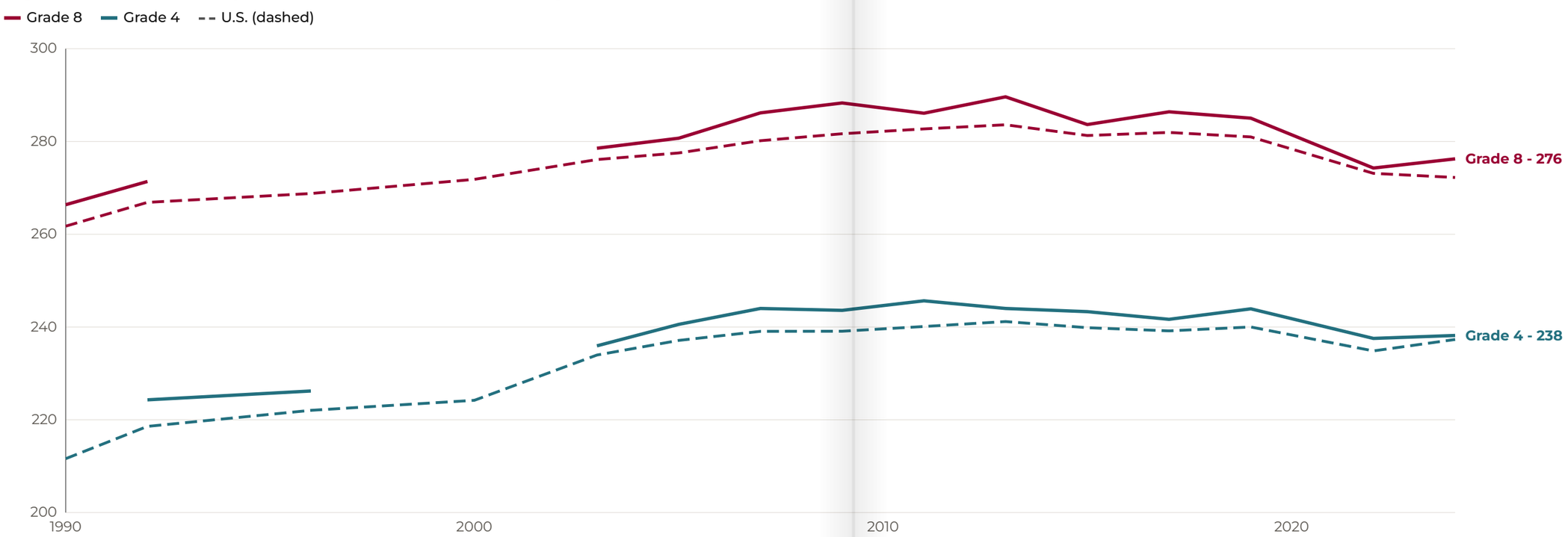
2.57K
Went to New York
The largest destination

SECTION 4

K-12 Preparation

Pennsylvania ranks 15th of 51 on the share of grade 8 mathematics: at or above Proficient, at 31% in 2024 against 27% for the U.S.

Figure 14 How Well Prepared in Mathematics Are Pennsylvania's Students?



Average NAEP mathematics score, public schools, 0–500 scale; state solid, U.S. public schools dashed. Data from NCES, National Assessment of Educational Progress.

The talent pipeline starts in school. Pennsylvania's eighth graders averaged 276 in mathematics on the 2024 NAEP, against 272 for U.S. public schools, 18th of 51 among the states.

31% scored at or above the Proficient level (U.S. 27%). Differences of a point or two are often within the margin of error.

276

Grade 8 math score (NAEP), 2024

U.S. **272** · **18th** of 51 · −7.4 since 2015

31%

Grade 8 math, at or above Proficient, 2024

U.S. **27%** · **15th** of 51 · −4.6 pp since 2015

259

Grade 8 reading score (NAEP), 2024

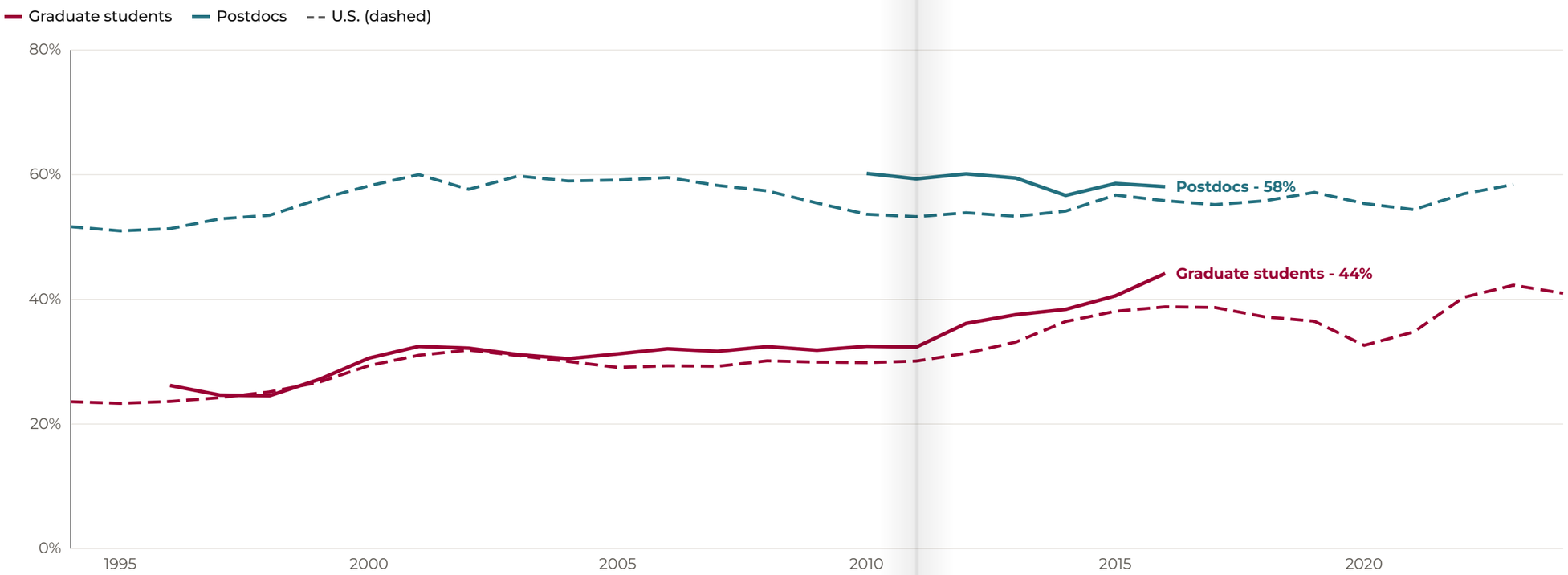
U.S. **257** · **19th** of 51 · −9.6 since 2015

SECTION 5

Graduate Research

Pennsylvania ranks 5th of 51 on research doctorates awarded, at 2.94K in 2025 against 59K for the U.S.

Figure 15 How Many of Pennsylvania's Science and Engineering Graduate Students Hold Temporary Visas?



Temporary-visa share of science and engineering graduate students and postdoctoral appointees, by the institution's state; state solid, U.S. dashed. NSF stopped publishing state tables after 2016. Data from NSF NCSES, Survey of Graduate Students and Postdoctorates.

In 2016, the last year NSF published state tables, 44% of Pennsylvania's science and engineering graduate students held temporary visas (U.S. 39%; 10th of 51).

Among postdoctoral researchers, the share was 58% (U.S. 56%). Postdocs are the stage of research training most dependent on international talent, and IPEDS does not count them.

The state's institutions awarded 2.94K research doctorates in 2025, 5.0% of the U.S. total (Data from NSF NCSES, Survey of Earned Doctorates; NSF publishes doctorates by citizenship only for the nation).

44%
S&E grad students on temporary visas, 2016
U.S. **39%** · **10th** of 51 · +3.6 pp since 2015

58%
S&E postdocs on temporary visas, 2016
U.S. **56%** · **18th** of 51 · −0.5 pp since 2015

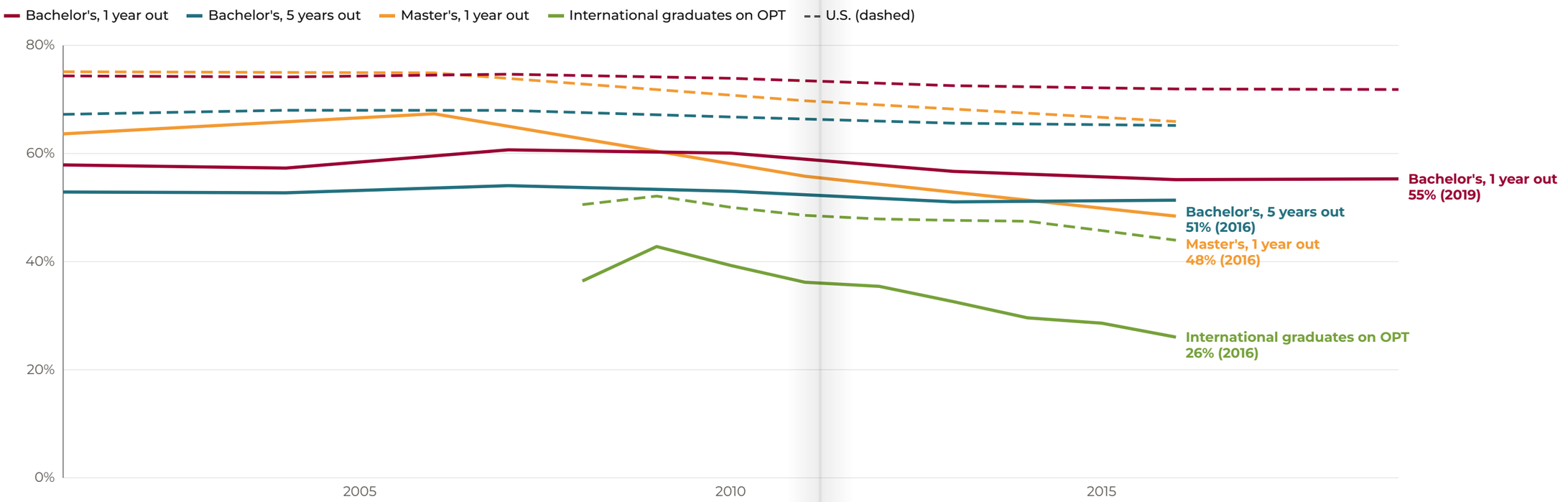
2.94K
Research doctorates awarded (NSF SED), 2025
U.S. **59K** · **5th** of 51 · +12% since 2015

SECTION 6

Do Graduates Stay?

Pennsylvania ranks 27th of 33 on the share of bachelor's graduates working in the state 1 year after graduating, at 55% in 2019 against 72% for the U.S.

Figure 16 How Many of Pennsylvania's Graduates Go On to Work in the State?



Share of employed graduates working in the state of their institution, by graduating cohort (Data from U.S. Census Bureau, Post-Secondary Employment Outcomes; participating institutions only); international graduates' OPT jobs in the state where they studied (Data from ICE, SEVIS FOIA releases; shares only, the file is truncated). State solid, U.S. or participating states dashed.

Of Pennsylvania's bachelor's graduates who were working a year after graduating, 55% worked in the state (participating states 72%). Five years out the share was 51%.

Among international graduates on Optional Practical Training in 2016, 26% worked in the state where they studied (U.S. 44%). That is the last year the released records cover, before most of the growth in international STEM master's degrees since 2015. Through migration between states, the state lost 2.6 bachelor's-plus adults per 1,000 such residents in 2025 (Data from U.S. Census Bureau, American Community Survey).

55%

Bachelor's grads working in-state, 1 year out, 2019

U.S. **72%** · **27th** of 33 · +0.2 pp since 2016

26%

OPT graduates working in the state, 2016

U.S. **44%** · **38th** of 51 · −2.6 pp since 2015

−2.6

Net migration, bachelor's-plus, per 1,000, 2025

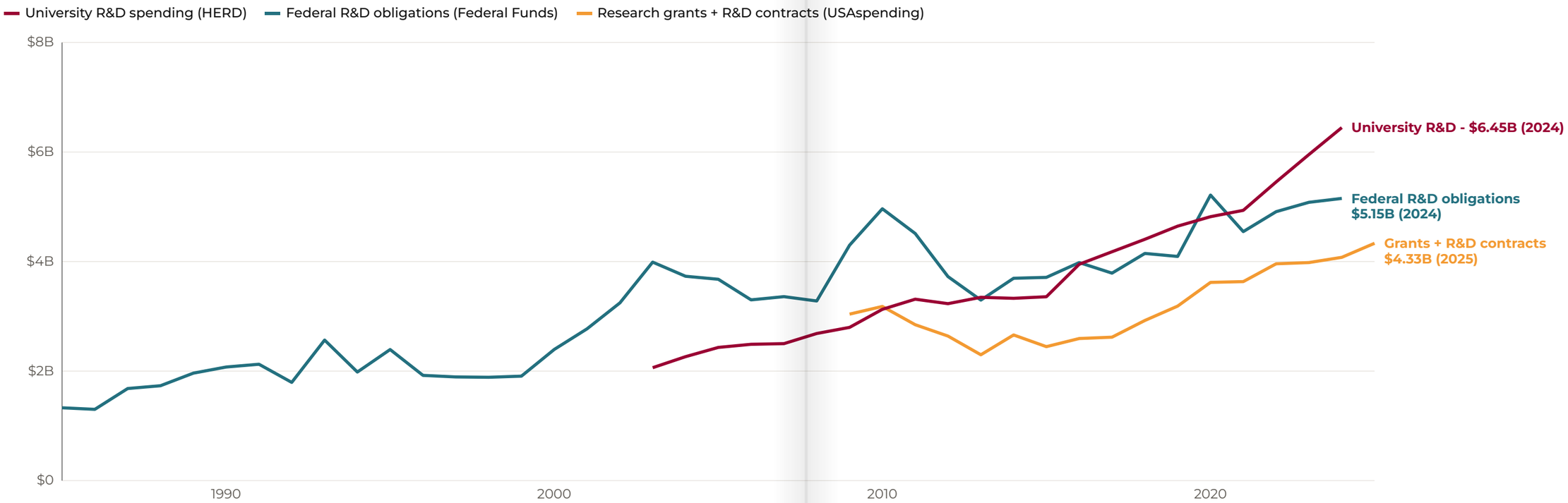
· **34th** of 51 · +2.3 since 2015

SECTION 7

Research Funding

Pennsylvania ranks 5th of 51 on university R&D spending, at $6.45B in 2024 against $118B for the U.S.

Figure 17 How Much Research Money Flows Into Pennsylvania?



Nominal dollars (not adjusted for inflation), fiscal years; a line ending before the others is labeled with its last year. University R&D: Data from NSF NCSES, Higher Education Research and Development Survey (FY2003-09 predecessor survey). Federal R&D obligations by state of performance: Data from NSF NCSES, Survey of Federal Funds for R&D (coverage widens in FY2021). Grants from NSF, NIH, NASA and DOE Science plus R&D contracts: Data from USAspending.gov.

Pennsylvania's universities spent $6.45B on research and development in FY2024, 5th of 51 among the states; 56% of it was federally funded (U.S. 55%).

Federal agencies obligated $5.15B for R&D performed in the state in FY2024. Research funding drives demand for graduate students and postdocs, the stage of training most dependent on international talent.

$6.45B

University R&D spending, 2024

U.S. **$118B** · **5th** of 51 · +92% since 2015

56%

Federal share of university R&D, 2024

U.S. **55%** · **17th** of 51 · −8.9 pp since 2015

$5.15B

Federal R&D obligations, 2024

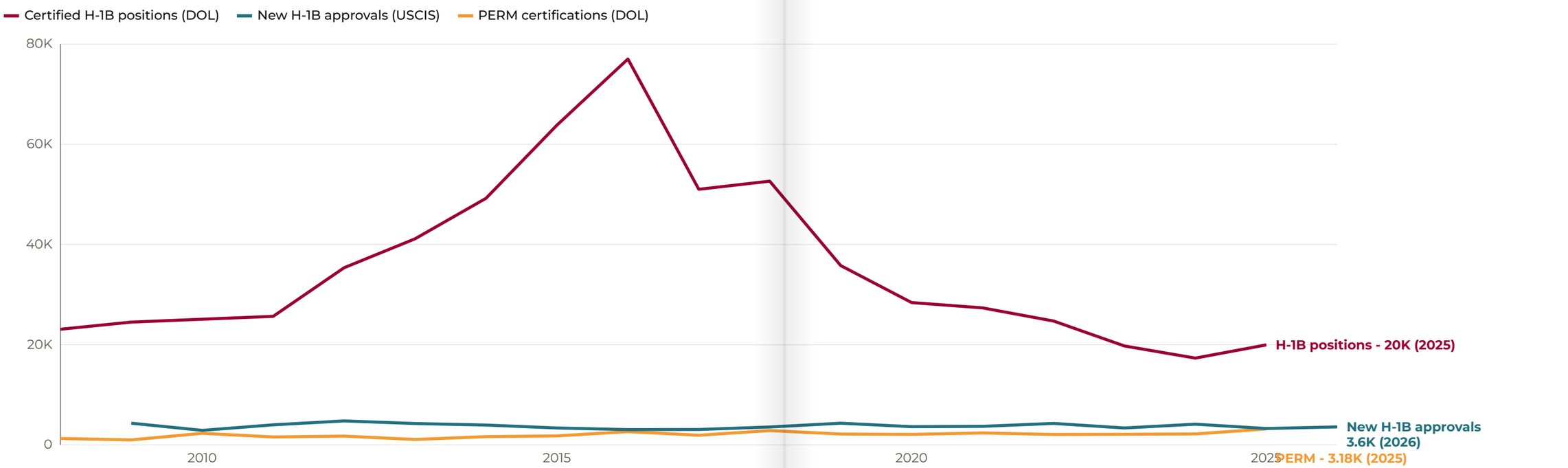
U.S. **$193B** · **12th** of 51 · +39% since 2015

SECTION 8

H-1B Jobs and Green Cards

Pennsylvania ranks 12th of 51 on certified H-1B positions, by worksite, at 20K in 2025 against 914K for the U.S.

Figure 18 How Many Skilled-Visa Jobs and Green-Card Certifications Does Pennsylvania Generate?



Certified H-1B positions on labor condition applications by worksite and certified PERM green-card applications, by fiscal year (Data from DOL, Office of Foreign Labor Certification); new H-1B petition approvals by the employer's address (Data from USCIS, H-1B Employer Data Hub).

Employers were certified for 20K H-1B positions at Pennsylvania worksites in FY2025, 12th of 51 among the states. A labor condition application comes before the petition and covers renewals and job changes as well as new hires, by the worksite's state, so it runs well above approvals. USCIS approved 3.6K new H-1B petitions in FY2026 and 8.55K continuing ones, counted by the employer's address.

DOL certified 3.18K PERM applications, the employer's step toward an employment-based green card, for jobs in the state in FY2025. In FY2024, 9.9% were for workers still on F-1 student visas (U.S. 8.7%): the most direct link from classroom to permanent residence.

20K

Certified H-1B positions (worksite), 2025

U.S. **914K** · **12th** of 51 · −69% since 2015

3.18K

PERM green-card certifications, 2025

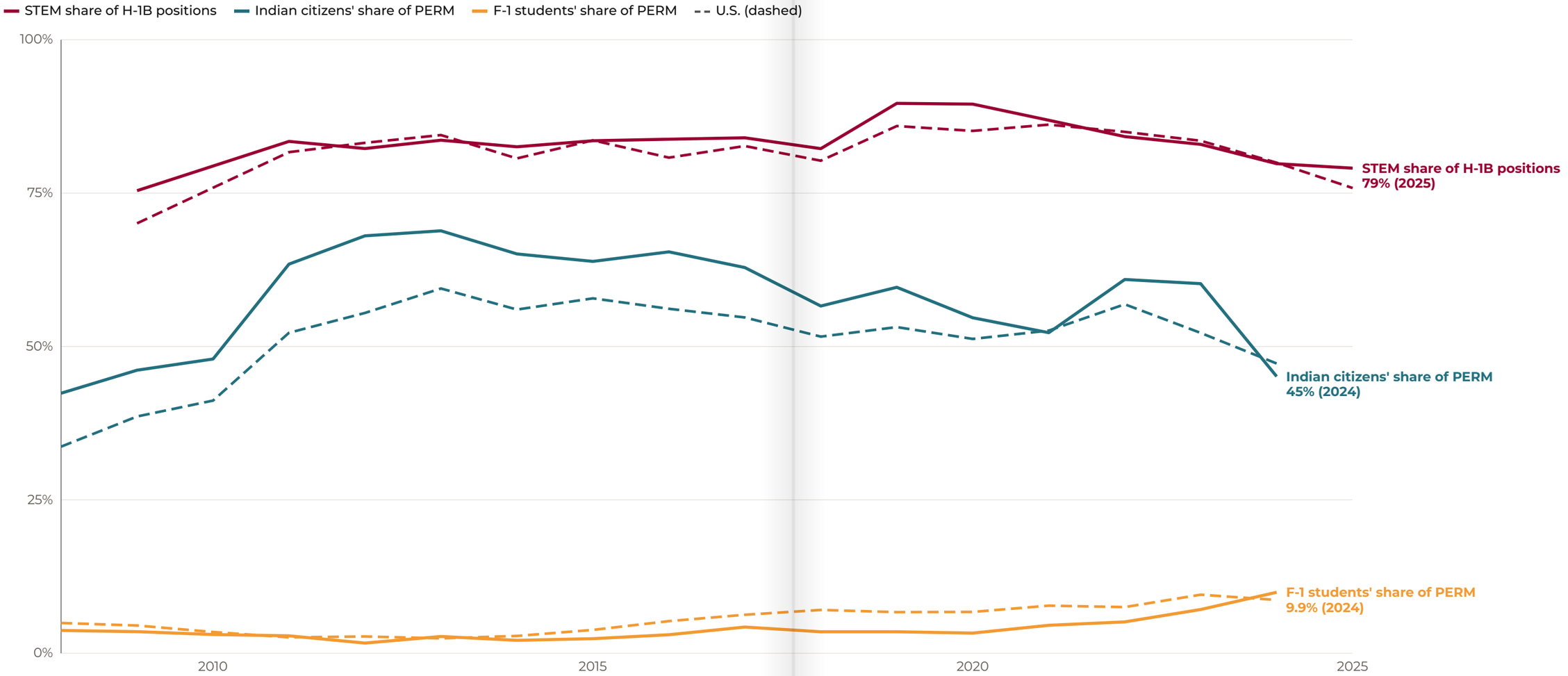
U.S. **138K** · **13th** of 51 · +77% since 2015

9.9%

PERM workers on F-1 student visas, 2024

U.S. **8.7%** · **24th** of 51 · +7.6 pp since 2015

Figure 19 Who Holds Pennsylvania's Skilled-Visa Jobs?



State solid, U.S. dashed. PERM citizenship and class of admission are not on the form DOL uses from FY2025. Data from DOL, Office of Foreign Labor Certification.

79% of the state's certified H-1B positions were in STEM occupations in FY2025 (U.S. 76%).

Citizens of India held 45% of the state's PERM certifications in FY2024 (U.S. 47%). Under per-country caps they face the longest waits for a green card, so this share measures the state's exposure to the backlog.

79%

STEM share of H-1B positions, 2025

U.S. **76%** · **11th** of 51 · −4.5 pp since 2015

45%

PERM certifications for Indian citizens, 2024

U.S. **47%** · **19th** of 51 · −18.7 pp since 2015

$103K

Median offered H-1B wage (nominal), 2025

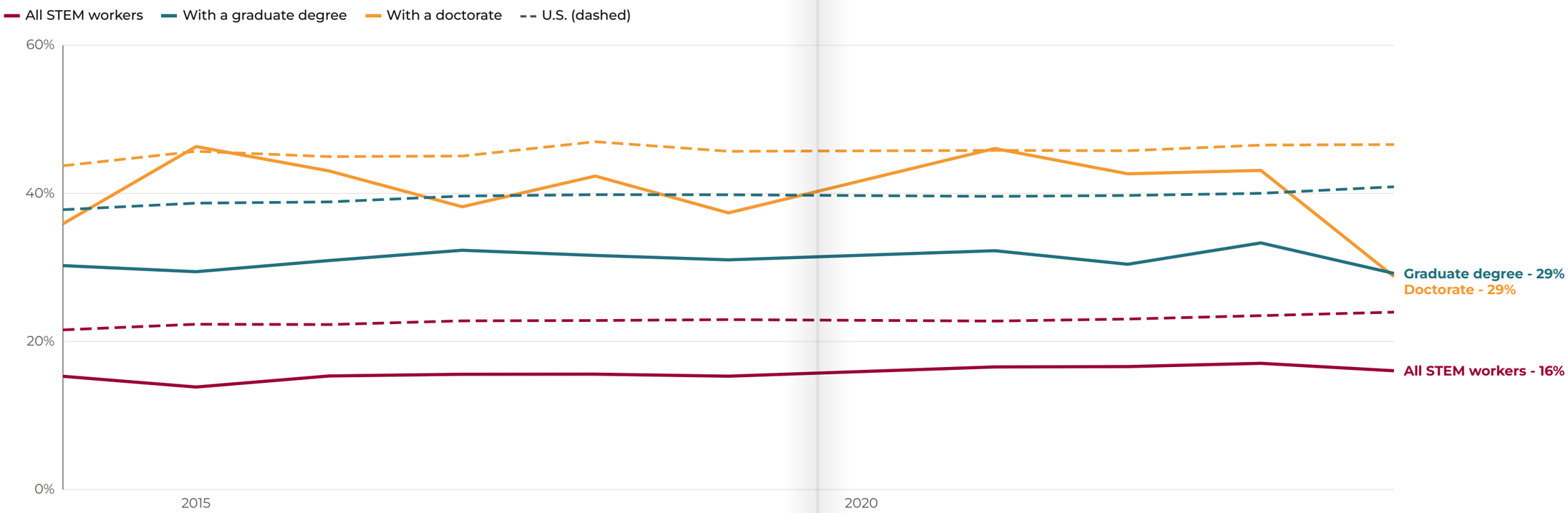
U.S. **$119K** · **27th** of 51 · +49% since 2015

SECTION 9

STEM Workforce

Pennsylvania ranks 21st of 51 on the foreign-born share of STEM workers, at 16% in 2024 against 24% for the U.S.

Figure 20 How Much of Pennsylvania's STEM Workforce Was Born Abroad?



Foreign-born share of employed workers in STEM occupations, by state of residence; state solid, U.S. dashed. Shares resting on fewer than 100 survey records are left out. No 2020 (experimental data). Data from U.S. Census Bureau, American Community Survey.

In 2024, 16% of the people working in STEM occupations in Pennsylvania were born abroad (U.S. 24%; 21st of 51).

The share rises with education: 29% of STEM workers with a graduate degree were foreign-born (U.S. 41%), and 29% of those with a doctorate (U.S. 47%; 19th of 20). 7.5% of all STEM workers were not U.S. citizens.

16%

Foreign-born share of STEM workers, 2024

U.S. **24%** · **21st** of 51 · +2.2 pp since 2015

29%

Foreign-born, STEM workers with grad degrees, 2024

U.S. **41%** · **23rd** of 44 · −0.2 pp since 2015

7.5%

Noncitizen share of STEM workers, 2024

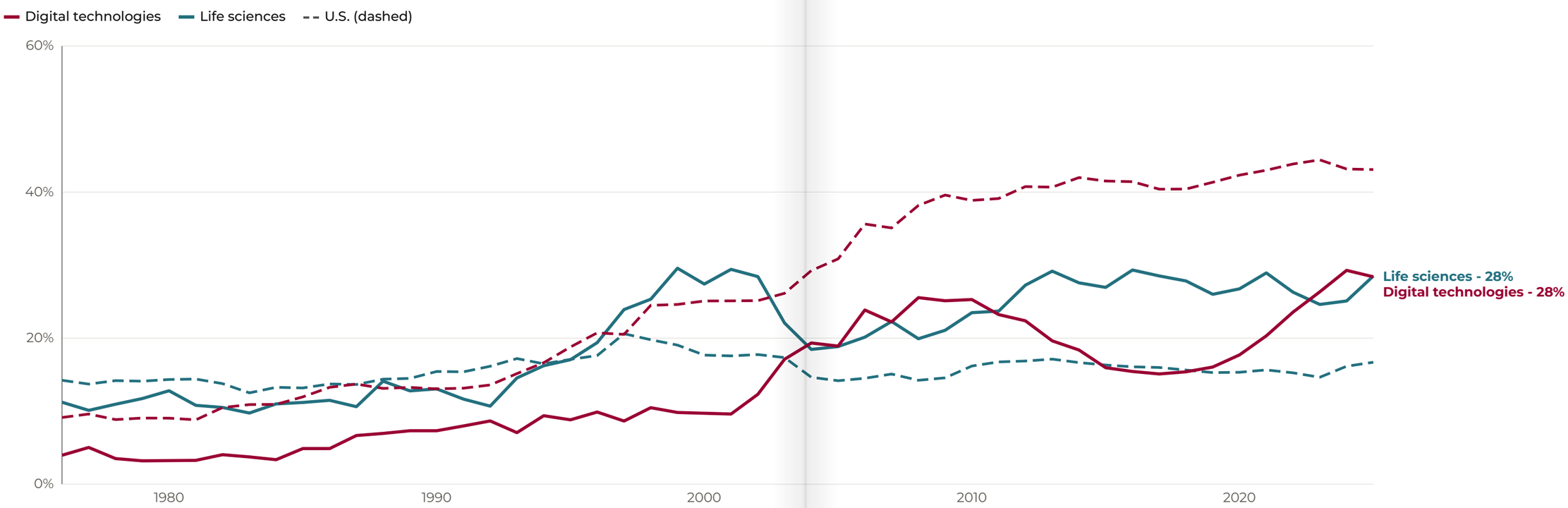
U.S. **11%** · **22nd** of 51 · +0.3 pp since 2015

SECTION 10

Patents

Pennsylvania ranks 10th of 51 on the share of the state's patents in life sciences, at 28% in 2025 against 17% for the U.S.

Figure 21 What Kinds of Technology Does Pennsylvania Patent?



Share of patents granted to owners in the state, by WIPO technology field of the first classification; state solid, U.S. dashed. Fractional counts by the owner's (assignee's) location, not where inventors live. Data from USPTO, PatentsView.

Organizations and people in Pennsylvania were granted 3.2K U.S. patents in 2025, 2.4% of all patents with a U.S. owner (12th of 51).

28% were in digital technologies (U.S. 43%) and 28% in the life sciences (U.S. 17%).

3.2K

Patents granted to owners in the state, 2025

U.S. **135K** · **12th** of 51 · +16% since 2015

28%

Patents in digital technologies, 2025

U.S. **43%** · **17th** of 51 · +12.5 pp since 2015

28%

Patents in life sciences, 2025

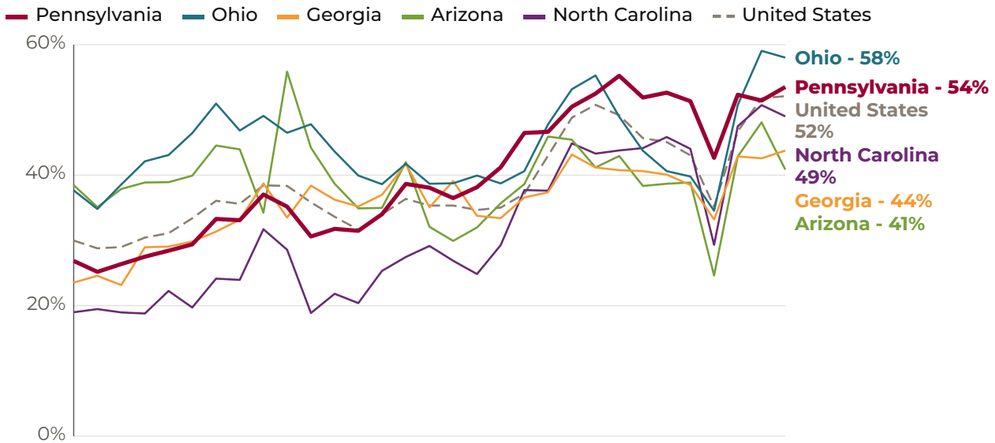
U.S. **17%** · **10th** of 51 · +1.6 pp since 2015

SECTION 11

Competitors

The four states closest in total fall enrollment, measured proportionally (a state 1.5 times as large counts as near as one two-thirds the size), are Ohio, Georgia, Arizona, and North Carolina. Pennsylvania ranks 2nd of the five on the international share of STEM master's; Ohio is the most dependent at 58%.

Figure 22 How Dependent Are Pennsylvania and Its Peers on International STEM Master's?



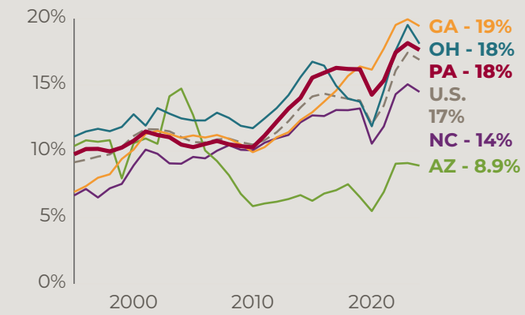
Data from NCES, IPEDS Completions.

Table 4 How Does Pennsylvania Compare With States of Similar Size?

Measure	PA	OH	GA	AZ	NC	U.S.
Students enrolled, 2024	**637K**	611K	594K	684K	582K	19.7M
Graduate share of students, 2024	**23%**	14%	18%	18%	14%	17%
STEM share of bachelor's, 2025	**27%**	24%	27%	17%	26%	24%
STEM share of master's, 2025	**25%**	27%	28%	15%	22%	23%
International STEM master's, 2025	**5.51K**	3.84K	3.78K	2.58K	2.42K	117K
Share of U.S. international STEM master's, 2025	**4.7%**	3.3%	3.2%	2.2%	2.1%	–
Int. share of STEM master's, 2025	**54%**	58%	44%	41%	49%	52%
Int. share of STEM research doctorates, 2025	**45%**	45%	44%	42%	39%	43%
Int. share of grad students, 2024	**18%**	18%	19%	8.9%	14%	17%
Int. share of undergraduate enrollment, 2024	**3.9%**	3.3%	2.6%	1.8%	2.1%	3.0%
Six-year graduation rate, 2024	**68%**	62%	57%	58%	66%	65%
In-state tuition, public 4-year, 2023	**$17K**	$11K	$7.14K	$10K	$7.54K	$9.71K
Appropriations per student, 2024	**$7.14K**	$7.76K	$10.8K	$9.12K	$15.6K	$11.4K
Average faculty salary, 2024	**$109K**	$93.3K	$95.3K	$102K	$95.1K	$106K
First-time students from in state, 2024	**68%**	76%	82%	61%	79%	–
Net migration per 100 residents, 2024	**+8**	+5	−6	+37	+4	–

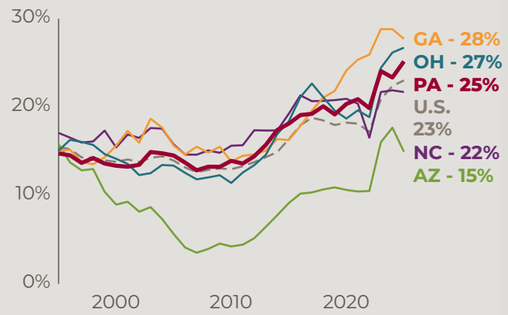
Latest year for each measure, given after its name. Dollars are 2024 dollars (CPI-U). Data from NCES, IPEDS.

Figure 23 Int. share of grad students



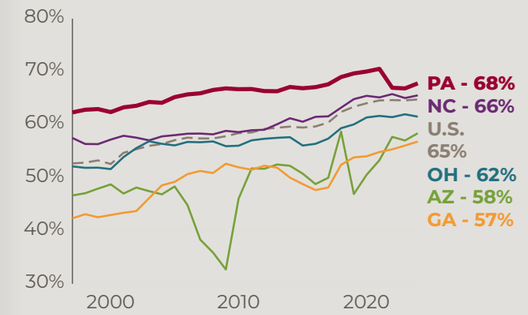
Data from NCES, IPEDS.

Figure 24 STEM share of master's



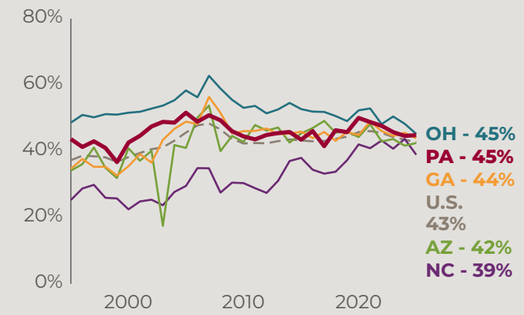
Data from NCES, IPEDS.

Figure 25 Six-year graduation rate



Data from NCES, IPEDS.

Figure 26 Int. share of STEM doctorates

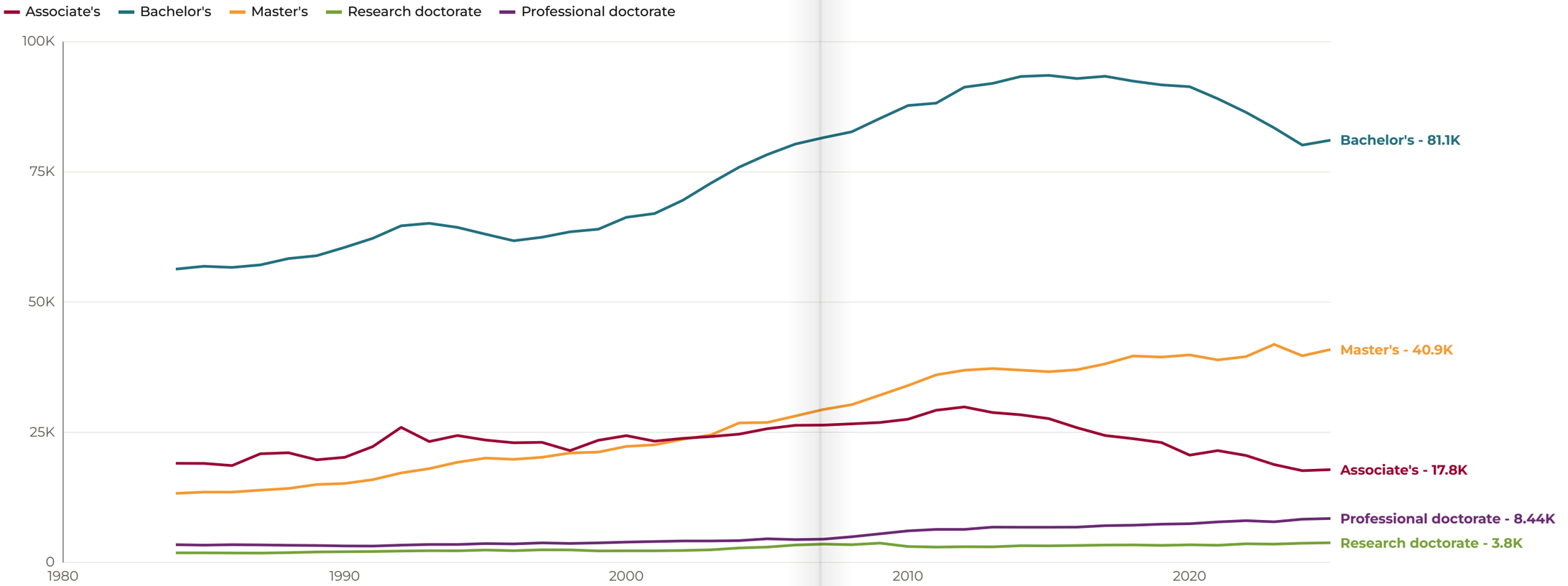


Data from NCES, IPEDS.

SECTION 12

Fields of Study

Figure 27 How Many Degrees Does Pennsylvania Award?



Degrees awarded by level, first majors. Data from NCES, IPEDS Completions.

In 2025, Pennsylvania's institutions awarded 17.8K associate's degrees, 81.1K bachelor's degrees, 40.9K master's degrees, 3.8K research doctorate degrees, 8.44K professional doctorate degrees.

Since 2015: associate's down 35%; bachelor's down 13%; master's up 12%; research doctorate up 19%; professional doctorate up 25%. The state awards 16.5 bachelor's degrees for every 100 undergraduates (U.S. 12.3).

Figure 28 What Do Pennsylvania's Bachelor's Graduates Study?



Share of bachelor's degrees by field, 2025. Data from NCES, IPEDS.

Pennsylvania's largest bachelor's fields are business (17%; U.S. 19%), health professions (13%; U.S. 13%), engineering (7.3%; U.S. 5.9%).

Relative to the nation, Pennsylvania's bachelor's degrees are most concentrated in physical sciences (1.50× the U.S. share), math & statistics (1.30× the U.S. share), engineering (1.24× the U.S. share).

6.9%

Bachelor's in computer science, 2025

U.S. **6.9%** · **18th** of 51 · +3.5 pp since 2015

8.1%

Bachelor's in engineering & eng. technology, 2025

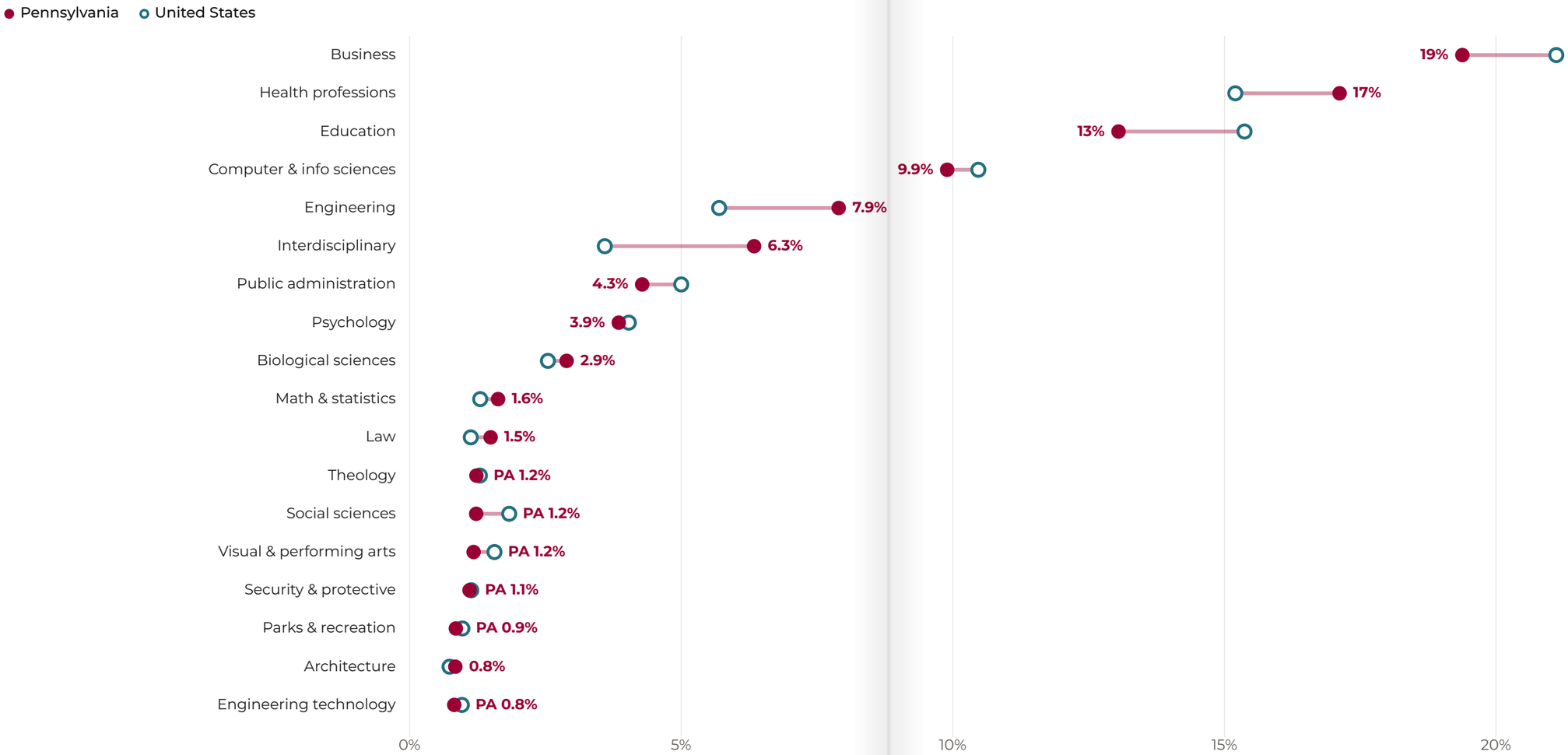
U.S. **6.8%** · **13th** of 51 · +1.1 pp since 2015

17%

Bachelor's in business, 2025

U.S. **19%** · **39th** of 51 · −1.2 pp since 2015

Figure 29 What Do Pennsylvania's Master's Graduates Study?



Share of master's degrees by field, 2025; largest fields. Data from NCES, IPEDS.

Pennsylvania's largest master's fields are business (19%; U.S. 21%), health professions (17%; U.S. 15%), education (13%; U.S. 15%).

International students earn 24% of all master's degrees in the state (U.S. 22%), and STEM fields account for 25% of them.

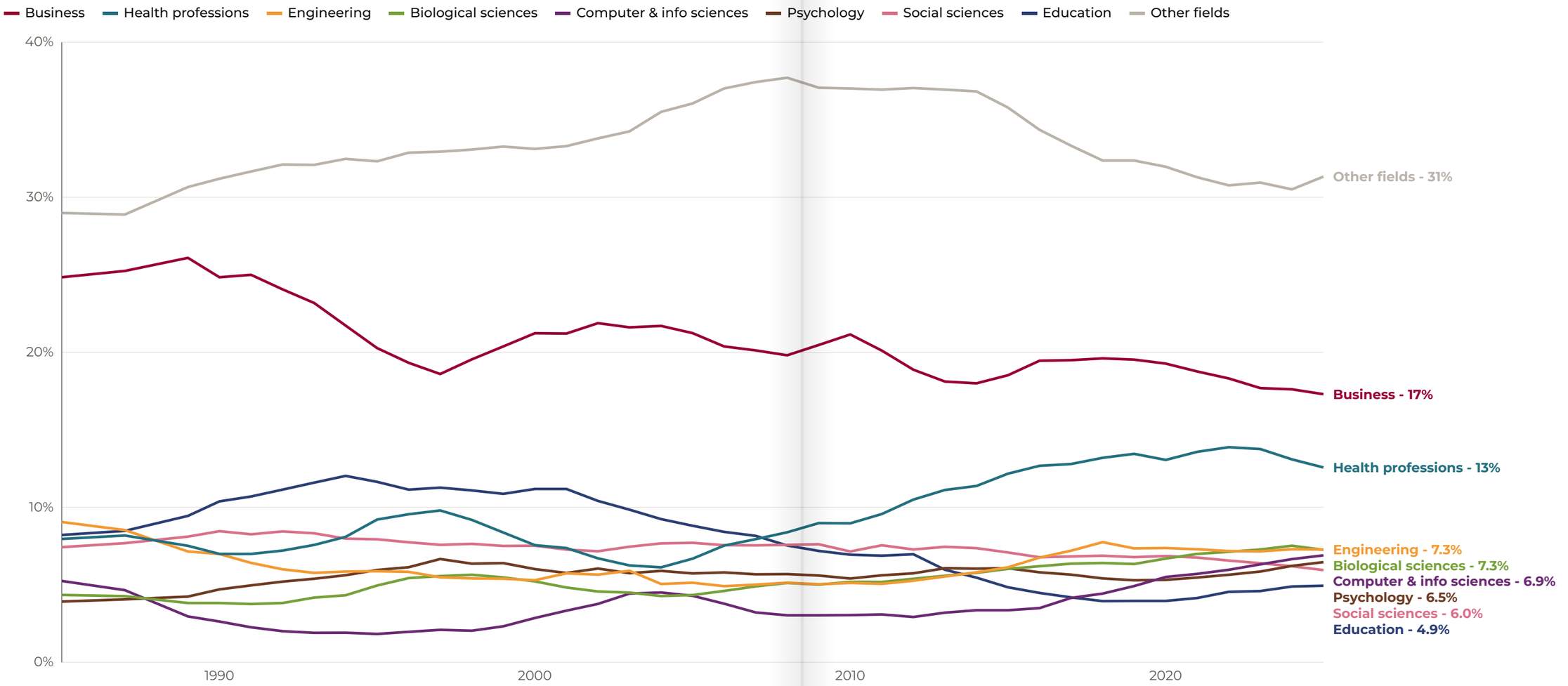
25%
STEM share of master's, 2025
U.S. **23%** · **15th** of 51 · +7.0 pp since 2015

24%
Int. share of all master's, 2025
U.S. **22%** · **12th** of 51 · +7.2 pp since 2015

62%
Int. share of CS master's, 2025
U.S. **65%** · **27th** of 47 · −1.1 pp since 2015

12

Figure 30 How Has Pennsylvania's Bachelor's Field Mix Changed?



Share of bachelor's degrees in the largest fields. Data from NCES, IPEDS Completions.

Since 2015, computer & info sciences has gained the most ground, 3.5 points of the state's bachelor's degrees, while business lost the most (1.2 points).

The chart follows the share of bachelor's degrees in each of the state's largest fields; smaller fields are grouped as other fields.

SECTION 12 · FIELDS OF STUDY

Where Is Pennsylvania Specialized, and What Do Its Doctoral Graduates Study?

The location quotient divides a field's share of Pennsylvania's bachelor's degrees by its share of all U.S. bachelor's degrees in 2025. A quotient of 1.00 means the same share as the nation; 2.00 means twice the national share. The table lists the 10 fields with the highest quotients among fields with at least 50 degrees.

Highest location quotients. Physical sciences (1.50, with 1.9% of the state's bachelor's degrees against 1.3% for the U.S.), math & statistics (1.30, with 1.5% of the state's bachelor's degrees against 1.2% for the U.S.), and engineering (1.24, with 7.3% of the state's bachelor's degrees against 5.9% for the U.S.) had the highest quotients.

Table 5 Where Is Pennsylvania Most Specialized?

Field	Location quotient	State share	U.S. share	Degrees
Physical sciences	**1.50**	1.9%	1.3%	1.56K
Math & statistics	**1.30**	1.5%	1.2%	1.24K
Engineering	**1.24**	7.3%	5.9%	5.9K
Architecture	**1.23**	0.7%	0.6%	564
Philosophy & religion	**1.21**	0.7%	0.6%	549
Interdisciplinary	**1.17**	3.4%	2.9%	2.72K
English	**1.15**	1.7%	1.5%	1.4K
History	**1.14**	1.2%	1.0%	956
Education	**1.14**	4.9%	4.3%	4.01K
Biological sciences	**1.10**	7.3%	6.6%	5.89K

Bachelor's degrees; location quotient = state share ÷ U.S. share; fields with 50+ degrees. Data from NCES, IPEDS.

27%
STEM share of bachelor's, 2025
U.S. **24%** · **10th** of 51

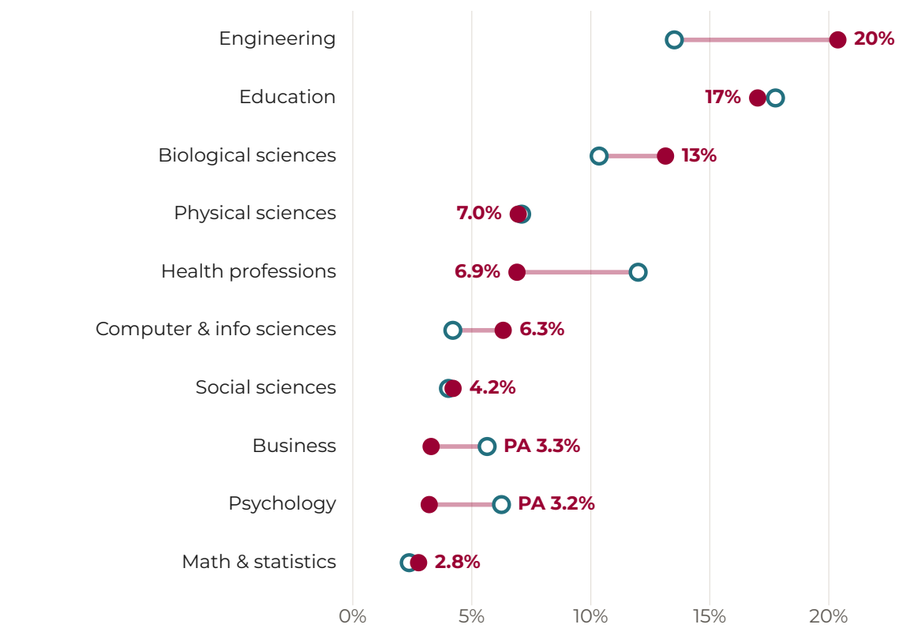
6.9%
Bachelor's in computer science, 2025
U.S. **6.9%** · **18th** of 51

8.1%
Bachelor's in engineering & eng. technology, 2025
U.S. **6.8%** · **13th** of 51

Each row is a field's share of the research doctorates awarded in Pennsylvania in 2025 (filled dot), against its share nationally (open dot), for the ten largest fields with at least 20 doctorates. Research doctorates exclude professional doctorates such as the M.D. and J.D.

Largest doctoral fields. The state's largest doctoral fields were engineering (20% of research doctorates, against 14% for the U.S.), education (17% of research doctorates, against 18% for the U.S.), and biological sciences (13% of research doctorates, against 10% for the U.S.).

Figure 31 What Do Pennsylvania's Doctoral Graduates Study?



Share of research doctorates by field, 2025; fields with 20+ doctorates. Data from NCES, IPEDS.

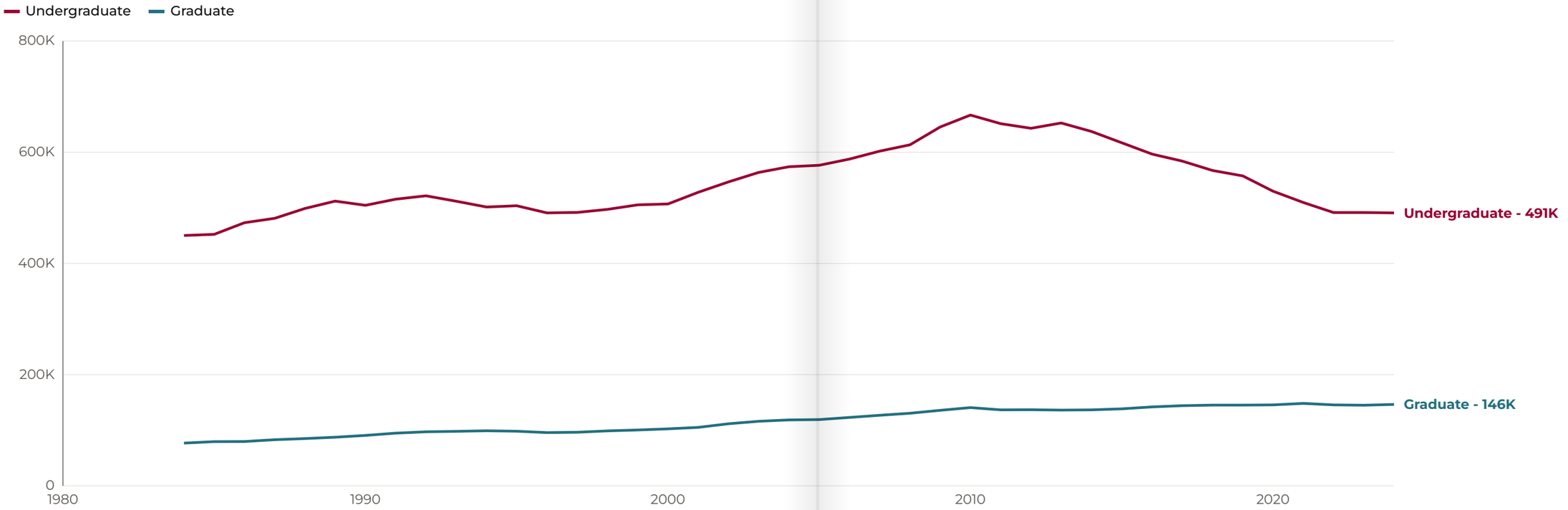
12

SECTION 13

Students and Money

Pennsylvania ranks 4th of 51 on the share of students who are graduate students, at 23% in 2024 against 17% for the U.S.

Figure 32 How Has Enrollment Changed?



Fall headcount enrollment. Data from NCES, IPEDS Fall Enrollment.

In fall 2024, Pennsylvania enrolled 491K undergraduates and 146K graduate students, 637K in all (7th of 51).

Undergraduate enrollment is down 20% since 2015 and graduate enrollment up 6% since 2015. Graduate students are 23% of all students (U.S. 17%).

637K

Students enrolled, 2024

U.S. **19.7M** · **7th** of 51 · −16% since 2015

23%

Graduate share of students, 2024

U.S. **17%** · **4th** of 51 · +4.6 pp since 2015

27%

Students attending part-time, 2024

U.S. **39%** · **48th** of 51 · −2.1 pp since 2015

SECTION 13 · STUDENTS AND MONEY

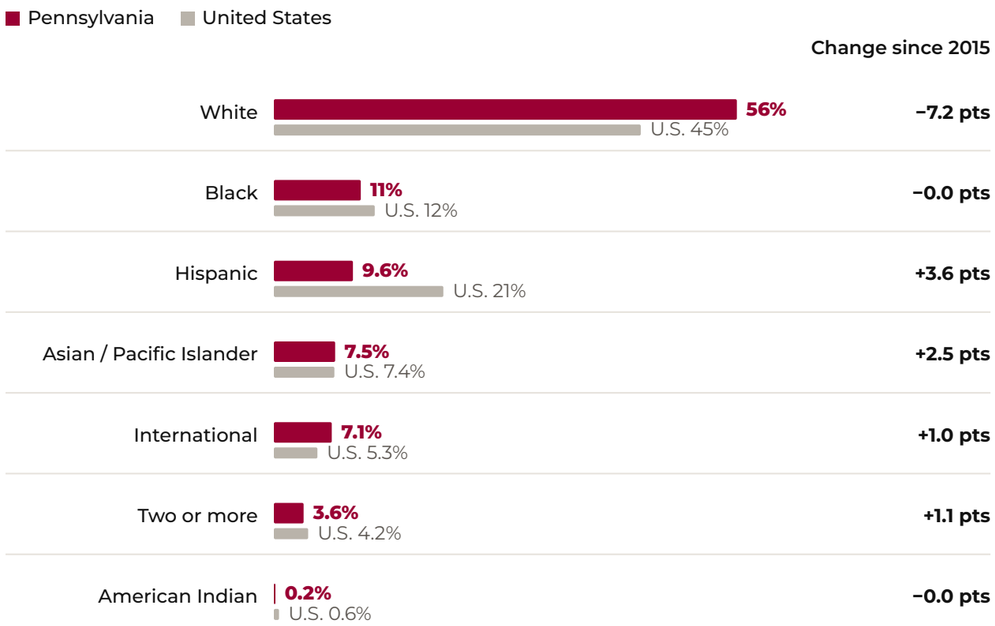
Who Enrolls in Pennsylvania, and Where?

In fall 2024, White students were the largest group at Pennsylvania's degree-granting institutions, **56%** of enrollment against 45% nationally, down 7.2 points since 2015.

Black students were **11%** (U.S. 12%), unchanged since 2015, and Hispanic students were **9.6%** (U.S. 21%), up 3.6 points since 2015.

Asian and Pacific Islander students were **7.5%** (U.S. 7.4%), up 2.5 points since 2015, and International students were **7.1%** (U.S. 5.3%), up 1.0 point since 2015. "International" counts U.S. nonresidents on temporary visas, whatever their race.

Figure 33 Who Enrolls in Pennsylvania?



Share of fall 2024 headcount enrollment by race and ethnicity; students of unknown race are left out of the chart. Data from NCES, IPEDS.

56%
White students
U.S. **45%** · −7.2 pp since 2015

11%
Black students
U.S. **12%** · −0.0 pp since 2015

45%
At private nonprofit institutions
+3.3 pp since 2015

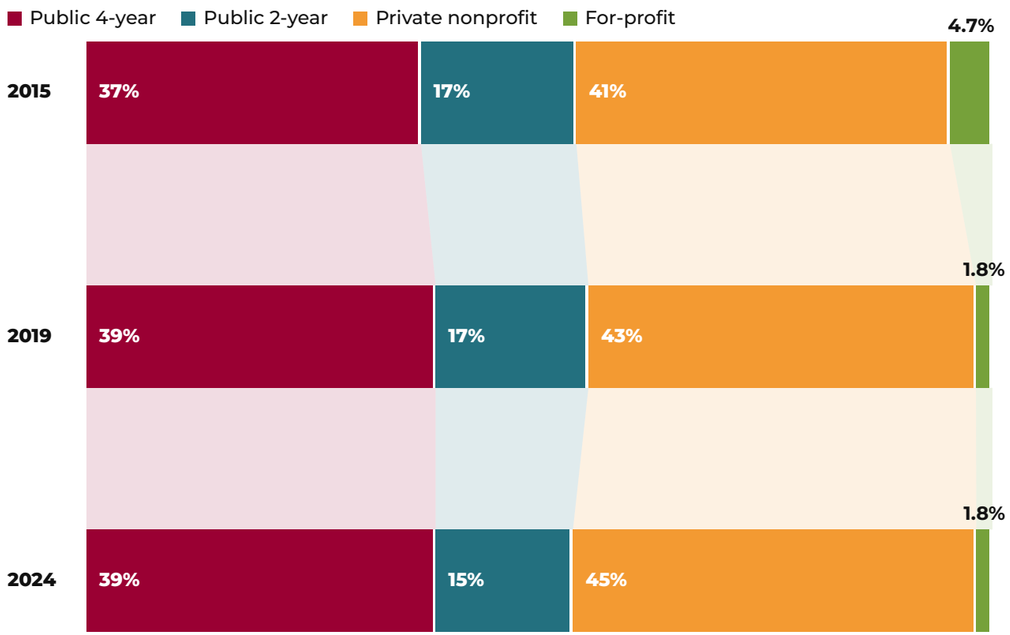
39%
At public 4-year institutions
+1.6 pp since 2015

In fall 2024, private nonprofit institutions enrolled **45%** of students, up 3.3 points from 41% in 2015. Public 4-year institutions enrolled **39%** of students, up 1.6 points from 37% in 2015.

Public 2-year institutions enrolled **15%** of students, down 2.0 points from 17% in 2015, and for-profit institutions enrolled **1.8%** of students, down 2.9 points from 4.7% in 2015.

Public institutions together enrolled **54%** of students, against 73% nationally, 43rd among the states.

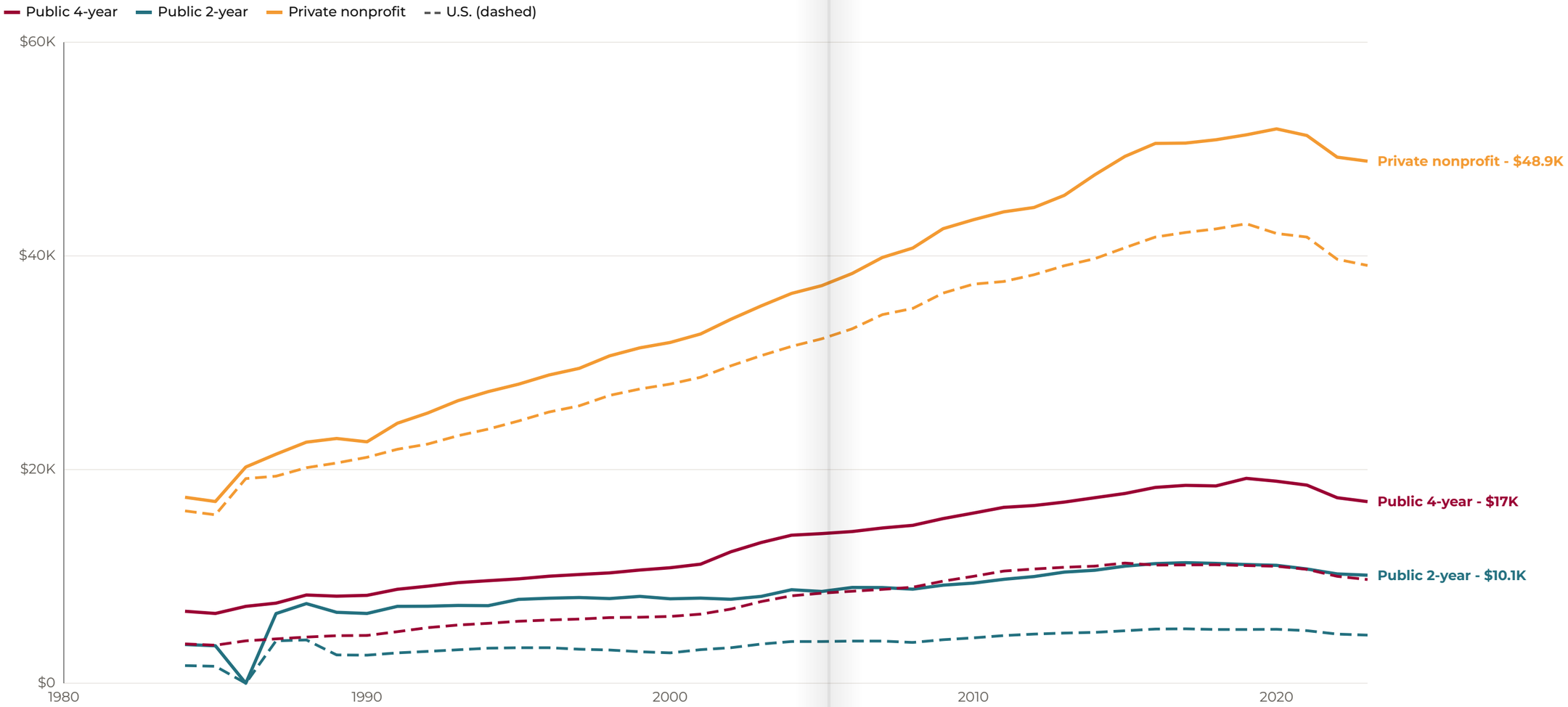
Figure 34 Where Do Students Enroll?



Share of fall headcount enrollment by sector, 2015, 2019, 2024. Sectors as IPEDS classifies them each year. Data from NCES, IPEDS.

13

Figure 35 What Does In-State Tuition Cost?



Published in-state tuition and fees, 2024 dollars (CPI-U); state solid, U.S. dashed. Data from NCES, IPEDS.

In-state tuition and fees at Pennsylvania's public four-year institutions are $17K (2023, 2024 dollars), 75% above the U.S. ($9.71K); the state ranks 3rd of 51.

State and local appropriations come to $7.14K per full-time-equivalent student at public institutions (46th of 51), and net tuition makes up 72% of net tuition plus state and local appropriations (U.S. 43%).

$17K

In-state tuition, public 4-year, 2023

U.S. **$9.71K** · **3rd** of 51 · −4.3% since 2015

$7.14K

Appropriations per student, 2024

U.S. **$11.4K** · **46th** of 51 · +20% since 2015

72%

Tuition share of tuition plus appropriations, 2024

U.S. **43%** · **3rd** of 51 · −4.0 pp since 2015

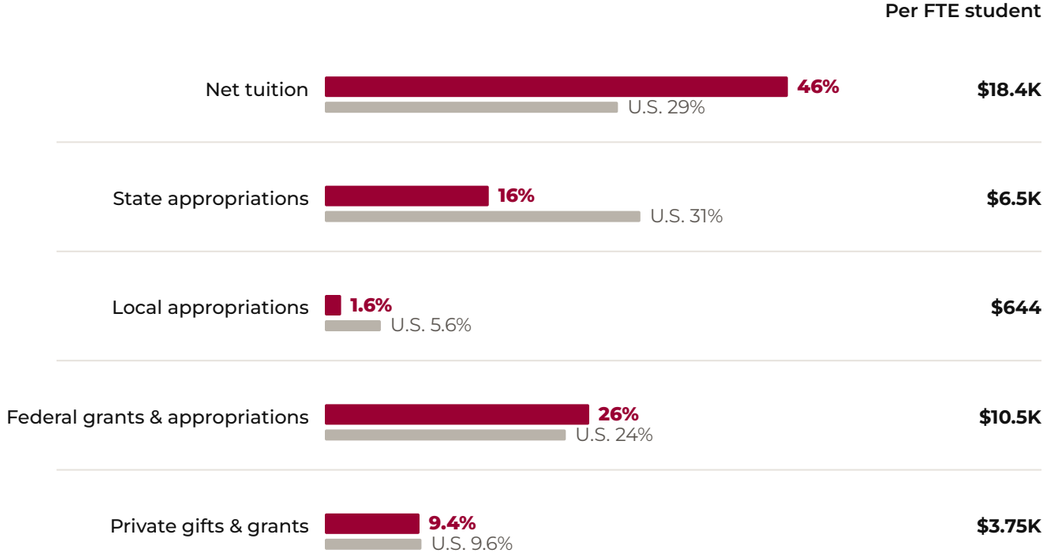
SECTION 13 · STUDENTS AND MONEY

How Are Pennsylvania's Public Institutions Funded, and What Do They Produce?

The bars give each source's share of the core revenue of Pennsylvania's public institutions, with the nation for comparison, and at right the dollars per full-time-equivalent (FTE) student it provides. Values are in 2024 dollars (CPI-U). Net tuition is tuition and fees after institutional discounts and allowances.

Net tuition per FTE student. The figure was $18.4K in FY2024, against $8.96K for the U.S., down 0.6% from $18.5K in FY2023 and down 2.9% since FY2015, when it was $18.9K.

Figure 36 How Are Public Institutions Funded?



Share of core revenue per FTE student by source, public institutions, FY2024; dollars per FTE at right, 2024 dollars (CPI-U). Shares are of all five sources; the tuition share elsewhere in this profile divides net tuition by net tuition plus state and local appropriations only. Data from NCES, IPEDS Finance.

$17K
In-state tuition, public 4-year, 2023
U.S. **$9.71K** · **3rd** of 51

$7.14K
Appropriations per student, 2024
U.S. **$11.4K** · **46th** of 51

72%
Tuition share of tuition plus appropriations, 2024
U.S. **43%** · **3rd** of 51

68%
Six-year graduation rate, 2024
U.S. **65%** · **15th** of 51

The table brings together five measures of what public higher education costs and what it produces in Pennsylvania: in-state tuition at public four-year institutions, state and local appropriations per full-time-equivalent student, tuition's share of public institutions' core revenue, the six-year bachelor's graduation rate, and the average full-time faculty salary.

In-state tuition, public 4-year. The figure was $17K in 2023, against $9.71K for the U.S., 3rd among the states, down 2.1% from $17.4K in 2022 and down 4.3% since 2015, when it was $17.8K.

Table 6 Money and Results

Measure	Year	Pennsylvania	U.S.	Rank	Change since 2015
In-state tuition, public 4-year	2023	**$17K**	$9.71K	3rd of 51	−4.3%
Appropriations per student	2024	**$7.14K**	$11.4K	46th of 51	+20%
Tuition share of tuition plus appropriations	2024	**72%**	43%	3rd of 51	−4.0 pp
Six-year graduation rate	2024	**68%**	65%	15th of 51	+0.9 pp
Average faculty salary	2024	**$109K**	$106K	14th of 51	−8.2%

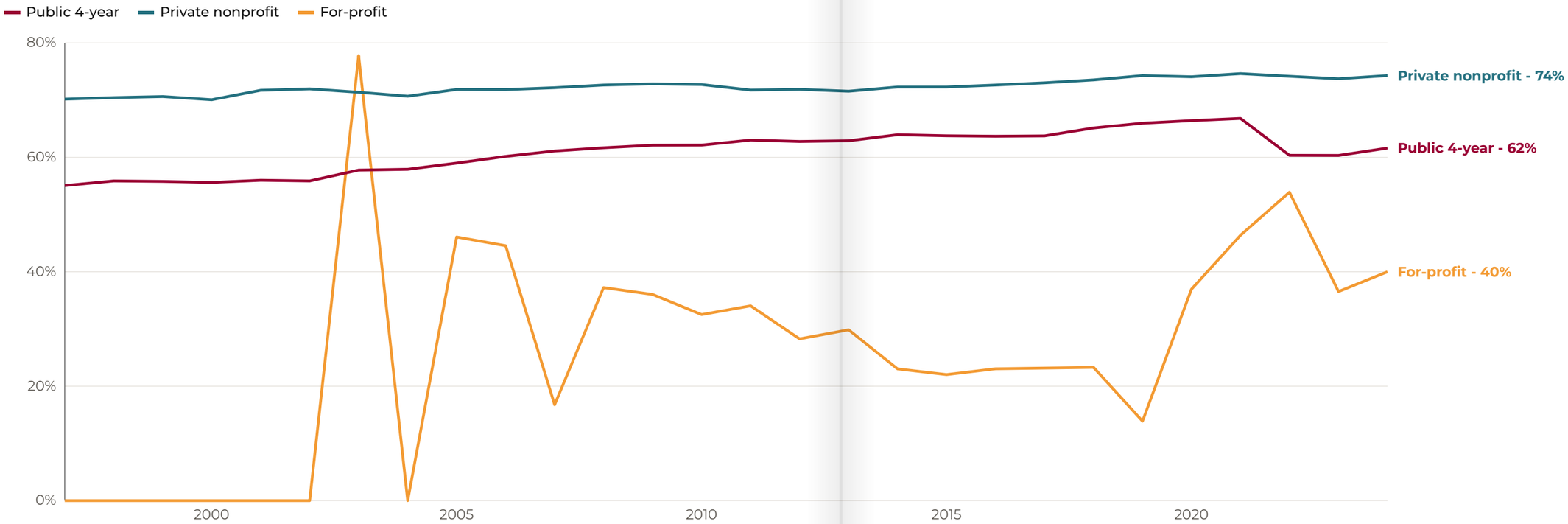
Rank among the 50 states and DC (1 = highest). Dollars in 2024 dollars (CPI-U). Data from NCES, IPEDS.

SECTION 14

Student Outcomes

Pennsylvania ranks 6th of 51 on bachelor's degrees per 100 undergraduates, at 16.5 in 2025 against 12.3 for the U.S.

Figure 37 How Do Graduation Rates Differ by Sector?



Six-year bachelor's graduation rate by sector of institution. Data from NCES, IPEDS Graduation Rates.

68% of students who started a bachelor's degree at a four-year institution in Pennsylvania graduated within six years (2024), 3.0 points above the U.S. (65%); the state ranks 15th of 51.

By sector: public 4-year 62%, private nonprofit 74%, for-profit 40%. At two-year institutions, 42% complete within 150% of normal time (U.S. 41%).

68%
Six-year graduation rate, 2024
U.S. **65%** · **15th** of 51 · +0.9 pp since 2015

42%
Two-year completion rate, 2024
U.S. **41%** · **22nd** of 50 · +5.4 pp since 2015

16.5
Bachelor's per 100 undergraduates, 2025
U.S. **12.3** · **6th** of 51 · +1.8 since 2015

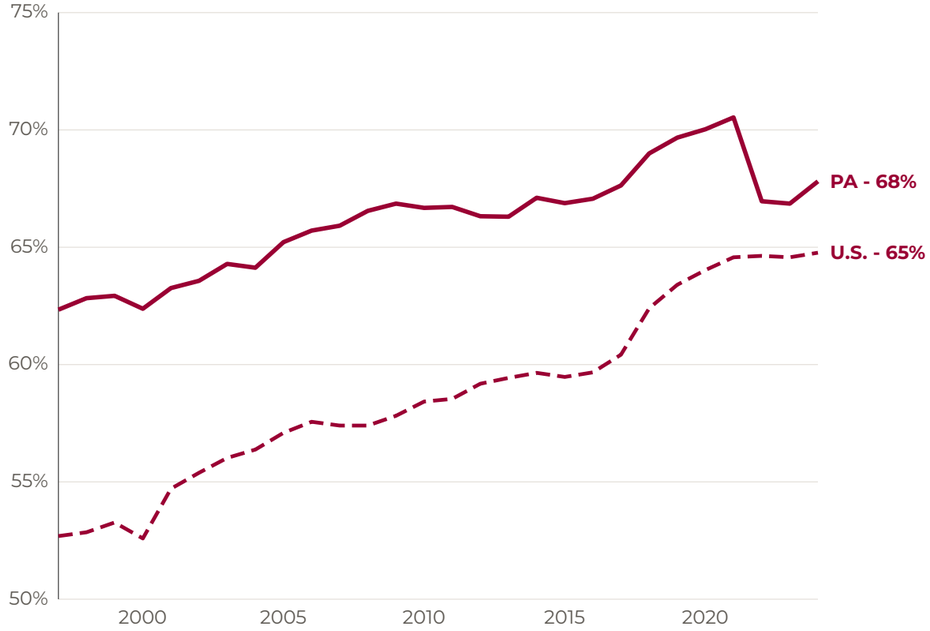
SECTION 14 · STUDENT OUTCOMES

How Many of Pennsylvania's Students Graduate?

The line follows the share of first-time, full-time bachelor's-degree students at Pennsylvania's four-year institutions who graduated from the institution where they started within six years (150% of normal time). Each year is the cohort that entered six years earlier. The rate leaves out students who start part-time (27% of the state's students attend part-time, against 39% nationally) and students who transfer and finish elsewhere. The state is solid and the United States dashed.

Six-year graduation rate. The rate was 68% in 2024, against 65% for the U.S., up 1.0 point from 67% in 2023 and up 0.9 points since 2015, when it was 67%.

Figure 38 Six-Year Bachelor's Graduation Rate



Four-year institutions; state solid, U.S. dashed. Data from NCES, IPEDS Graduation Rates.

68%
Six-year graduation rate, 2024
U.S. **65%** · **15th** of 51

42%
Two-year completion rate, 2024
U.S. **41%** · **22nd** of 50

16.5
Bachelor's per 100 undergraduates, 2025
U.S. **12.3** · **6th** of 51

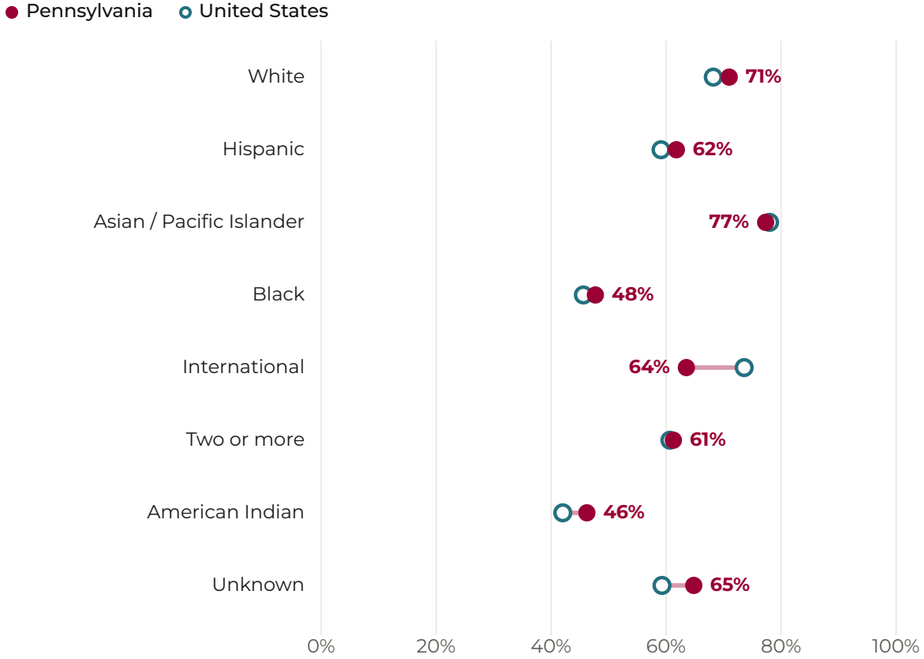
Each row is the six-year bachelor's graduation rate for one racial or ethnic group of students at Pennsylvania's four-year institutions (filled dot), against the national rate for the same group (open dot). Groups with fewer than 100 students in the cohort are left out.

White. The six-year rate was 71% in 2024, against 68% for the U.S.

Hispanic. The six-year rate was 62% in 2024, against 59% for the U.S.

Asian / Pacific Islander. The six-year rate was 77% in 2024, against 78% for the U.S.

Figure 39 How Do Graduation Rates Differ by Race and Ethnicity?



Six-year rate, 2024; groups with 100+ students. Data from NCES, IPEDS.

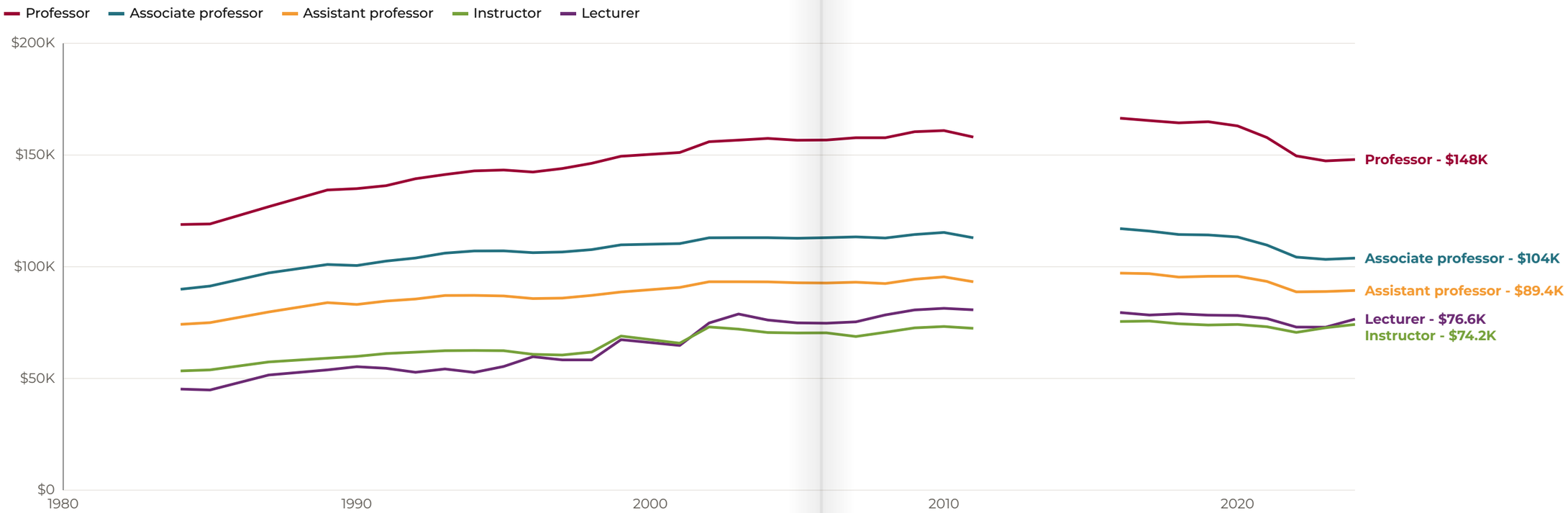
14

SECTION 15

Faculty

Pennsylvania ranks 14th of 51 on the average full-time faculty salary, at $109K in 2024 against $106K for the U.S.

Figure 40 What Do Pennsylvania's Faculty Earn, by Rank?



Average salary of full-time instructional faculty on 9/10-month contracts, by academic rank, 2024 dollars (CPI-U). Data from NCES, IPEDS Human Resources.

Full-time instructional faculty on 9/10-month contracts at Pennsylvania's institutions earned an average of $109K in 2024 (2024 dollars), 3% above the U.S. ($106K); the state ranks 14th of 51. By rank (2024): professors $148K, associate professors $104K, assistant professors $89.4K, instructors $74.2K, lecturers $76.6K.

Pennsylvania's institutions had 34.6K full-time instructional staff in 2024, down 6% since 2015. Average salary across all ranks is down 8% since 2016.

$109K

Average faculty salary, 2024

U.S. **$106K** · **14th** of 51 · −8.2% since 2016

SECTION 15 · FACULTY

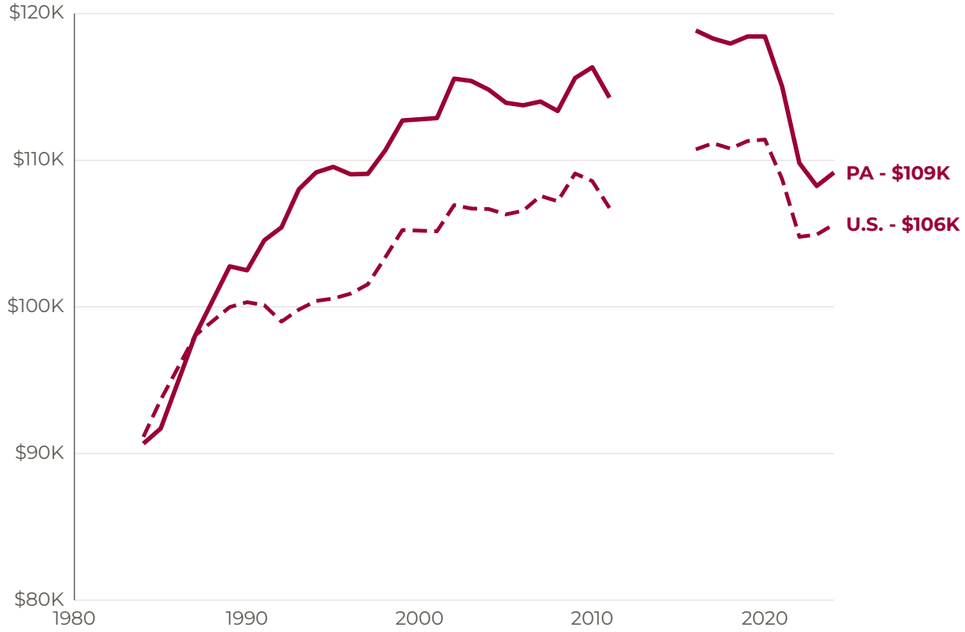
What Do Pennsylvania's Faculty Earn, and How Many Hold Tenure?

The line follows the average salary of full-time instructional faculty on 9- or 10-month contracts at Pennsylvania's degree-granting institutions, all ranks combined, in 2024 dollars (CPI-U). The state is solid and the United States dashed.

Average salary. The figure was $109K in 2024, against $106K for the U.S., 14th among the states, up 0.9% from $108K in 2023 and down 8.2% since 2016, when it was $119K.

Full-time instructional staff. The number was 34.6K in 2024, down 1.7% from 35.2K in 2023 and down 5.7% since 2015, when it was 36.7K.

Figure 41 Average Faculty Salary



Full-time instructional faculty, 9/10-month, 2024 dollars. Data from NCES, IPEDS.

$109K
Average faculty salary, 2024
U.S. **$106K** · **14th** of 51

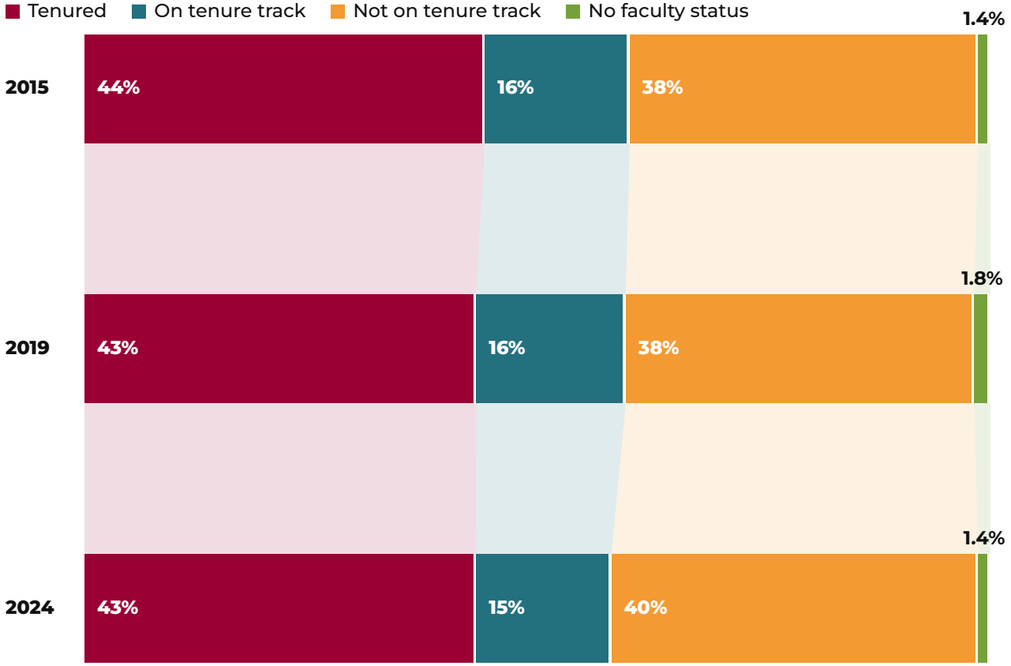
34.6K
Full-time instructional staff, 2024

43%
Tenured, 2024

The bars divide Pennsylvania's full-time instructional staff by tenure status in 2015, 2019 and 2024: tenured, on the tenure track, not on the tenure track (for example, lecturers on fixed-term contracts), and staff at institutions with no tenure system.

Tenured, share of full-time instructional staff. The share was 43% in 2024, up 0.3 points from 43% in 2023 and down 1.0 point since 2015, when it was 44%.

Figure 42 Tenure Status



Full-time instructional staff by tenure status, 2015, 2019, 2024. Data from NCES, IPEDS.

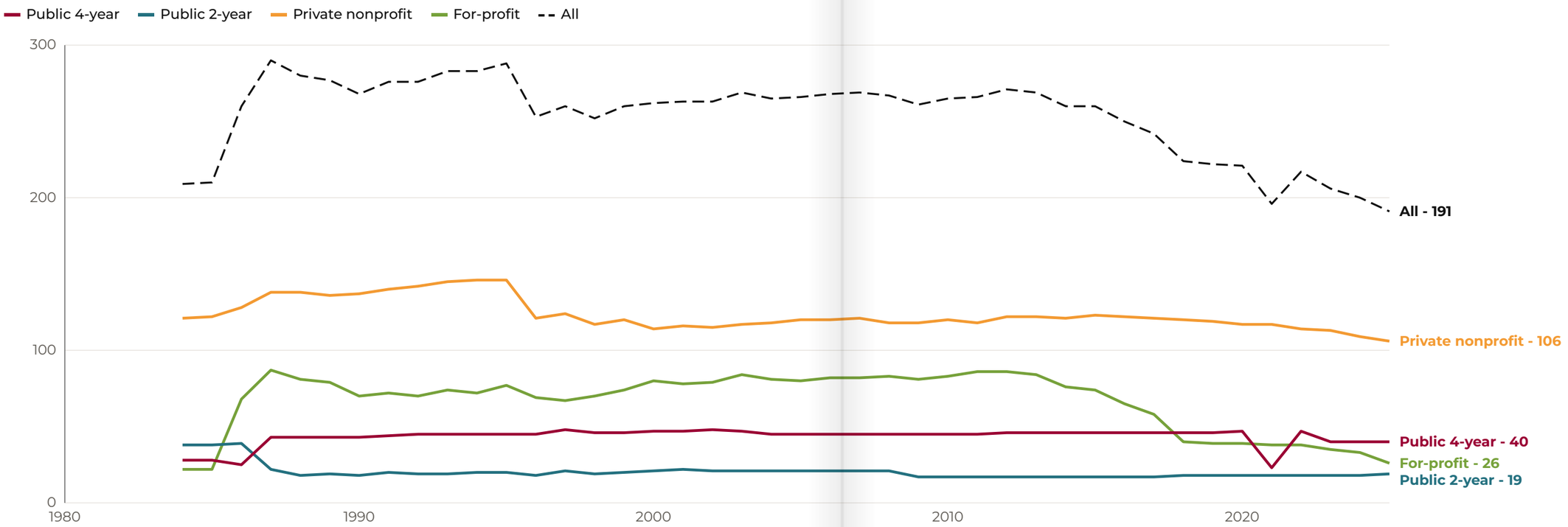
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SECTION 16

Institutions

Pennsylvania ranks 46th of 51 on the share of students enrolled exclusively online, at 17% in 2024 against 27% for the U.S.

Figure 43 How Many Institutions Does Pennsylvania Have?



Degree-granting institutions by sector. Data from NCES, IPEDS.

Pennsylvania has 191 degree-granting institutions:40 public 4-year, 19 public 2-year, 106 private nonprofit, 26 for-profit (2025).

The number of institutions is down 27% since 2015. 54% of students attend public institutions and 1.8% attend for-profit ones.

54%

Students at public institutions, 2024

U.S. **73%** · **43rd** of 51 · −0.4 pp since 2015

1.8%

Students at for-profit institutions, 2024

U.S. **5.5%** · **26th** of 51 · −2.9 pp since 2015

17%

Students fully online, 2024

U.S. **27%** · **46th** of 51 · +5.5 pp since 2015

SECTION 16 · INSTITUTIONS

Which Institutions Lead in Pennsylvania?

The table ranks institutions in Pennsylvania by the STEM master's degrees and research doctorates they awarded to international students in 2025, the two counted together. The share columns give international students' share of each institution's STEM degrees at that level.

Largest. Carnegie Mellon University (2.21K international STEM master's degrees and 193 doctorates), University of Pennsylvania (1.21K international STEM master's degrees and 122 doctorates), and Pennsylvania State University-Main Campus (319 international STEM master's degrees and 269 doctorates) awarded the most.

Table 7 Which Institutions Train the Most International STEM Graduates?

Institution	Intl STEM master's	Int. share	Intl STEM doctorates	Int. share
Carnegie Mellon University Pittsburgh · Private nonprofit	**2.21K**	74%	193	55%
University of Pennsylvania Philadelphia · Private nonprofit	**1.21K**	52%	122	35%
Pennsylvania State University-Main Campus University Park · Public 4-year	**319**	50%	269	51%
Harrisburg University of Science and Technology Harrisburg · Private nonprofit	**454**	97%	4	40%
University of Pittsburgh-Pittsburgh Campus Pittsburgh · Public 4-year	**236**	39%	132	46%
Drexel University Philadelphia · Private nonprofit	**258**	37%	39	27%
Gannon University Erie · Private nonprofit	**271**	95%	–	–
Robert Morris University Moon Township · Private nonprofit	**208**	89%	4	31%

Degrees awarded in 2025. Data from NCES, IPEDS.

191
Degree-granting institutions, 2025

54%
Students at public institutions, 2024
U.S. **73%** · **43rd** of 51

1.8%
Students at for-profit institutions, 2024
U.S. **5.5%** · **26th** of 51

17%
Students fully online, 2024
U.S. **27%** · **46th** of 51

The table lists Pennsylvania's largest degree-granting institutions by fall 2024 headcount enrollment, with the international share of graduate students, STEM bachelor's degrees awarded in 2025, the six-year graduation rate and the admission rate (a dash where the institution admits all applicants or does not report one).

Largest. The three largest were Pennsylvania State University-Main Campus (50.7K students), University of Pittsburgh-Pittsburgh Campus (35.7K students), and Temple University (29.6K students).

Table 8 What Are Pennsylvania's Largest Institutions?

Institution	Enrollment	Int. share, grad	STEM bachelor's	Six-year grad rate	Admit rate
Pennsylvania State University-Main Campus	**50.7K**	44%	4.16K	86%	61%
University of Pittsburgh-Pittsburgh Campus	**35.7K**	22%	2.2K	85%	58%
Temple University	**29.6K**	13%	1.14K	75%	80%
University of Pennsylvania	**29.1K**	30%	950	97%	5.4%
Drexel University	**21.2K**	15%	1.04K	78%	79%
West Chester University of Pennsylvania	**17.2K**	1.9%	263	71%	78%
Carnegie Mellon University	**15.9K**	62%	1.23K	94%	12%
Pennsylvania State University-World Campus	**13.8K**	3.5%	302	34%	91%
Community College of Philadelphia	**12.9K**	–	–	–	–
Harrisburg Area Community College	**12.6K**	–	–	–	–

Fall 2024 enrollment; degrees 2025. Data from NCES, IPEDS.

NOTES

About These Data

International students

U.S. nonresidents on temporary visas, as IPEDS defines them. IPEDS does not record country of citizenship. Permanent residents count with U.S. citizens.

STEM

CIP families 03, 04, 11, 14, 15, 26, 27 and 40. Doctorates are research and scholarship doctorates; professional doctorates appear only in the degree-count figure. Degrees are first majors only.

Universe

Degree-granting institutions located in the state; Title IV participants from 1996. Geography is the institution's location, not the student's home. Enrollment is fall headcount.

Migration

First-time degree-seeking undergraduates by home state, reported in even years before 2001 and every year since; odd years with incomplete reporting are omitted. Net migration is students arriving from other states minus residents leaving, per 100 resident first-time students.

Money

Tuition, appropriations, revenue and faculty salaries are in 2024 dollars using the CPI-U. Research funding (HERD, Federal Funds, USAspending), wages and the NAFSA economic contribution are nominal dollars, and are labeled so. Appropriations and revenue are per full-time-equivalent student at public institutions; the tuition share divides net tuition by net tuition plus state and local appropriations.

Race and ethnicity

IPEDS did not collect race and ethnicity in every year before 2010, and categories changed in 2008–2010; “Two or more” exists only from then.

Countries of citizenship

From DHS SEVIS by the Numbers: active F-1 and M-1 student records in the month named, at every level from primary school to doctorate, by the state of the school. IPEDS records only whether a student is a U.S. nonresident, not where they come from. SEVIS counts are monthly snapshots and are not directly comparable with IPEDS fall enrollment or degree counts. Year-to-year comparisons use one fall snapshot per year (the month closest to November).

Peers

Comparison states are the four closest in total fall enrollment, measured proportionally: a state 1.5 times as large counts as near as one two-thirds the size.

Sources

IPEDS (National Center for Education Statistics) is the backbone: Institutional Characteristics, Fall Enrollment, Completions, Graduation Rates, Finance, Human Resources and Admissions, 1980–2025, harmonized into a state panel. The profile also draws on NAEP, NSF's Survey of Graduate Students and Postdoctorates, Survey of Earned Doctorates, HERD and Federal Funds surveys, USAspending.gov, DHS SEVIS by the Numbers and ICE SEVIS releases, the Department of Labor's Office of Foreign Labor Certification and BLS wage data, the USCIS H-1B Employer Data Hub, the Census Bureau's American Community Survey and Post-Secondary Employment Outcomes, USPTO PatentsView, and other federal sources; every figure names its source, and the Sources page lists them in full. Generated November 2026.

NOTES

How the Numbers Are Counted

Ranks

Ranks are among the 50 states and the District of Columbia (1 = highest value, whether or not a high value is good). Where fewer than 51 report, the count is given ("23rd of 33").

Sectors

Sectors are as IPEDS classifies institutions each year. IPEDS moves a whole institution to "public, 4-year or above" once it awards any bachelor's degree, so community colleges that began offering applied bachelor's degrees carry all of their students into the public 4-year sector. Sector series (enrollment shares, public 4-year tuition, graduation rates by sector) reflect these moves.

Research doctorates

Two counts appear. IPEDS counts degrees institutions report as "doctor's degree, research/scholarship", which includes some practice-oriented degrees such as many Ed.D.s; the NSF Survey of Earned Doctorates counts research doctorate recipients and is smaller. Each figure names which it uses. "STEM master's per STEM research doctorate" divides IPEDS STEM master's degrees by IPEDS STEM research doctorates.

Engineering

"Engineering" alone is CIP 14. Where engineering technology (CIP 15) is added, the label says "engineering & eng. technology".

Graduation rates

IPEDS graduation rates follow first-time, full-time students and count only those who finish at the institution where they started, so they leave out part-time starters and students who transfer and finish elsewhere.

International students in SEVIS

SEVIS counts active F-1 and M-1 records, which include graduates on Optional Practical Training (up to 36 months for STEM graduates, 12 for others), so STEM shares run high. STEM follows the DHS STEM Designated Degree Program List, which is broader than the eight CIP families above, and DHS draws its STEM pages on a different basis from its all-student pages, so a country's STEM count can exceed its all-level count.

Visas and green cards

DOL labor condition applications (H-1B positions) are counted by worksite and include renewals and job changes; USCIS approvals are counted by the employer's address and split into new and continuing. PERM certifications are the employer's step toward a green card.

Patents

Patents are counted by the owner's (assignee's) location, not where inventors live; a company headquartered in another state takes its patents there.

Years

Each measure uses its latest year, so years differ across a page; every figure, table and statement gives its year, and lines that stop early are labeled with their last year.

Pennsylvania – By the Numbers

Measure	2015	Latest	Change since 2015	U.S., latest	U.S. change since 2015	Rank	Source
K-12 PREPARATION							
Grade 8 mathematics, average score	284	**276**[2024]	−7.4 ▼	272	−9.0 ▼	#18	[5]
Grade 8 mathematics: at or above Proficient	36%	**31%**[2024]	−4.6 pp ▼	27%	−5.3 pp ▼	#15	[5]
Grade 4 mathematics, average score	243	**238**[2024]	−5.1 ▼	237	−2.5 ▼	#23	[5]
Grade 4 mathematics: at or above Proficient	45%	**41%**[2024]	−3.6 pp ▼	39%	0.0 pp	#18	[5]
Grade 8 reading, average score	269	**259**[2024]	−9.6 ▼	257	−7.3 ▼	#19	[5]
Grade 8 reading: at or above Proficient	39%	**31%**[2024]	−8.3 pp ▼	29%	−4.0 pp ▼	#18	[5]
Grade 4 reading, average score	227	**216**[2024]	−10.9 ▼	214	−7.1 ▼	#17	[5]
Grade 8 science, average score, 2009 framework	–	**151**[2011]	–	151	–	#30	[5]
Current spending per public-school pupil	$18.8K	**$21K**[2021]	+11% ▲	$16.8K	+12% ▲	#13	[3]
Public high school 4-year graduation rate, ACGR	85%	**87%**[2022]	+2.0 pp ▲	87%	+4.0 pp ▲	#19	[3]
Public high school graduates	125K	**127K**[2024]	+1.8% ▲	3.49M	+8.8% ▲	#6	[3]
Average public-school teacher salary	$84.3K	**$79.3K**[2022]	−5.9% ▼	$72.9K	−3.0% ▼	#12	[3]
Pupils per teacher, public schools	14.2	**13.3**[2022]	−0.9 ▼	15.4	−0.6 ▼	#36	[3]
Public elementary and secondary enrollment	1.72M	**1.69M**[2024]	−1.5% ▼	49.4M	−2.1% ▼	#7	[3]
Public-school students eligible for free or reduced-price lunch	46%	**57%**[2023]	+11.0 pp ▲	53%	+1.5 pp ▲	#15	[3]
English learners as a share of public-school students	3.0%	**4.6%**[2022]	+1.6 pp ▲	11%	+1.0 pp ▲	#40	[3]
Charter-school share of public-school enrollment	7.7%	**9.7%**[2023]	+2.0 pp ▲	7.6%	+2.2 pp ▲	#12	[3]
High schools offering calculus	79%[2016]	**74%**[2024]	−5.5 pp ▼	51%	−5.8 pp ▼	#2	[14]
High schools offering physics	83%[2016]	**84%**[2024]	+0.2 pp ▲	64%	−2.4 pp ▼	#6	[14]
High schools offering chemistry	91%[2016]	**90%**[2024]	−0.9 pp ▼	81%	+0.7 pp ▲	#12	[14]
High schools offering Algebra II	93%[2016]	**91%**[2024]	−2.2 pp ▼	87%	+1.3 pp ▲	#21	[14]
High schools offering computer science	66%[2021]	**65%**[2024]	−1.5 pp ▼	55%	+2.0 pp ▲	#23	[14]
High schools offering any AP course	79%[2016]	**79%**[2024]	−0.2 pp ▼	56%	0.0 pp	#7	[14]
High schools offering AP computer science	28%[2021]	**37%**[2024]	+8.5 pp ▲	25%	+2.6 pp ▲	#12	[14]

Superscript years mark the year observed; changes since 2015 start from 2015 or the first later year with data. Rank is among the 50 states and DC (1 = highest). "pp" is percentage points. Dollars are 2024 dollars (CPI-U) unless the measure is labeled nominal. Counts and dollars are rounded to three significant figures (K, M, B). Arrow colors show direction only. Source numbers refer to the list of sources that follows the tables.

PENNSYLVANIA – BY THE NUMBERS (CONTINUED)

Measure	2015	Latest	Change since 2015	U.S., latest	U.S. change since 2015	Rank	Source
K-12 PREPARATION (CONTINUED)							
High schools offering dual enrollment	60%[2016]	**74%[2024]**	+14.1 pp ▲	71%	+11.1 pp ▲	#31	[14]
AP course enrollments per 100 high-school students	14.1[2016]	**14.2[2024]**	0.0	18.4	+1.0 ▲	#29	[14]
Calculus enrollments per 100 high-school students	5.6[2016]	**3.9[2024]**	−1.6 ▼	3.3	−1.0 ▼	#11	[14]
Physics enrollments per 100 high-school students	10.1[2016]	**8.1[2024]**	−2.0 ▼	9.4	−0.4 ▼	#28	[14]
SAT or ACT takers per 100 high-school students	19.2[2016]	**10.1[2024]**	−9.0 ▼	18.1	−2.5 ▼	#45	[14]
Students in gifted and talented programs per 100 students	3.6[2016]	**2.8[2024]**	−0.8 ▼	7.0	+0.4 ▲	#35	[14]
ENROLLMENT AND INSTITUTIONS							
Pell Grant share of undergraduates	32%	**34%[2025]**	+2.0 pp ▲	36%	−0.8 pp ▼	#31	[15]
Fall enrollment	755K	**637K[2024]**	−16% ▼	19.7M	−1.7% ▼	#7	[4]
Share of enrollment at public institutions	54%	**54%[2024]**	−0.4 pp ▼	73%	0.0 pp	#43	[4]
Share of enrollment at for-profit institutions	4.7%	**1.8%[2024]**	−2.9 pp ▼	5.5%	−1.2 pp ▼	#26	[4]
Share of students enrolled part-time	29%	**27%[2024]**	−2.1 pp ▼	39%	+0.7 pp ▲	#48	[4]
Share of students who are graduate students	18%	**23%[2024]**	+4.6 pp ▲	17%	+2.0 pp ▲	#4	[4]
Share of students enrolled exclusively online	11%	**17%[2024]**	+5.5 pp ▲	27%	+12.1 pp ▲	#46	[4]
DEGREES AND FIELDS OF STUDY							
Research doctorates awarded	2.63K	**2.94K[2025]**	+12% ▲	59K	+7.3% ▲	#5	[7]
Science & engineering research doctorates, excluding health	1.87K	**2.25K[2025]**	+21% ▲	44.4K	+14% ▲	#5	[7]
Share of U.S. research doctorates	4.8%	**5.0%[2025]**	+0.2 pp ▲	–	–	#5	[7]
Engineering research doctorates	618	**732[2025]**	+18% ▲	10.9K	+9.9% ▲	#3	[7]
Mathematics & computer science research doctorates	216	**279[2025]**	+29% ▲	5.07K	+33% ▲	#6	[7]
STEM master's degrees per research doctorate	4.2	**5.4[2025]**	+1.2 ▲	6.4	+2.1 ▲	#25	[4]
STEM share of bachelor's	21%	**27%[2025]**	+6.2 pp ▲	24%	+5.3 pp ▲	#10	[4]
Share of bachelor's in computer & information sciences	3.4%	**6.9%[2025]**	+3.5 pp ▲	6.9%	+3.7 pp ▲	#18	[4]
Share of bachelor's in engineering & eng. technology	6.9%	**8.1%[2025]**	+1.1 pp ▲	6.8%	+0.7 pp ▲	#13	[4]
Share of bachelor's in mathematics & statistics	1.4%	**1.5%[2025]**	+0.1 pp ▲	1.2%	0.0 pp	#8	[4]

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PENNSYLVANIA – BY THE NUMBERS (CONTINUED)

Measure	2015	Latest	Change since 2015	U.S., latest	U.S. change since 2015	Rank	Source
DEGREES AND FIELDS OF STUDY (CONTINUED)							
Share of bachelor's in biological & biomedical sciences	6.0%	**7.3%**[2025]	+1.2 pp ▲	6.6%	+0.8 pp ▲	#15	[4]
Share of bachelor's in physical sciences	2.2%	**1.9%**[2025]	−0.3 pp ▼	1.3%	−0.3 pp ▼	#4	[4]
Share of bachelor's in health professions	12%	**13%**[2025]	+0.4 pp ▲	12%	+1.1 pp ▲	#22	[4]
Share of bachelor's in business	19%	**17%**[2025]	−1.2 pp ▼	19%	−0.1 pp ▼	#39	[4]
STEM share of master's	18%	**25%**[2025]	+7.0 pp ▲	23%	+6.7 pp ▲	#15	[4]
Share of master's in computer & information sciences	5.0%	**9.9%**[2025]	+4.9 pp ▲	10%	+6.3 pp ▲	#17	[4]
Share of master's in engineering & eng. technology	8.0%	**8.7%**[2025]	+0.7 pp ▲	6.7%	−0.2 pp ▼	#8	[4]
Share of master's in mathematics & statistics	1.0%	**1.6%**[2025]	+0.7 pp ▲	1.3%	+0.3 pp ▲	#11	[4]
Share of master's in biological & biomedical sciences	2.1%	**2.9%**[2025]	+0.8 pp ▲	2.5%	+0.6 pp ▲	#19	[4]
Share of master's in physical sciences	0.9%	**0.7%**[2025]	−0.1 pp ▼	0.8%	−0.1 pp ▼	#28	[4]
Share of master's in health professions	17%	**17%**[2025]	+0.5 pp ▲	15%	+1.7 pp ▲	#13	[4]
Share of master's in business	22%	**19%**[2025]	−3.0 pp ▼	21%	−3.2 pp ▼	#32	[4]
STEM share of research doctorates	50%	**50%**[2025]	+0.3 pp ▲	39%	−2.2 pp ▼	#10	[4]
Share of research doctorates in computer & information sciences	4.9%	**6.3%**[2025]	+1.4 pp ▲	4.2%	+1.4 pp ▲	#8	[4]
Share of research doctorates in engineering & eng. technology	20%	**21%**[2025]	+0.5 pp ▲	14%	−1.0 pp ▼	#5	[4]
Share of research doctorates in mathematics & statistics	2.7%	**2.8%**[2025]	0.0 pp	2.4%	−0.2 pp ▼	#20	[4]
Share of research doctorates in biological & biomedical sciences	13%	**13%**[2025]	+0.1 pp ▲	10%	−1.1 pp ▼	#18	[4]
Share of research doctorates in physical sciences	8.2%	**7.0%**[2025]	−1.2 pp ▼	7.1%	−1.1 pp ▼	#33	[4]
Share of research doctorates in health professions	8.1%	**6.9%**[2025]	−1.2 pp ▼	12%	+3.5 pp ▲	#34	[4]
Share of research doctorates in business	2.8%	**3.3%**[2025]	+0.5 pp ▲	5.6%	+1.3 pp ▲	#28	[4]
Bachelor's degrees per 100 undergraduates	14.7	**16.5**[2025]	+1.8 ▲	12.3	+1.4 ▲	#6	[4]
INTERNATIONAL STUDENTS							
First-time students from outside the U.S.	3.5%[2016]	**3.4%**[2024]	−0.1 pp ▼	3.1%	+0.1 pp ▲	#16	[4]
STEM master's degrees earned by international students	3.1K	**5.51K**[2025]	+78% ▲	117K	2.18× ▲	#7	[4]
Share of all U.S. international STEM master's degrees	5.8%	**4.7%**[2025]	−1.1 pp ▼	–	–	#7	[4]

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PENNSYLVANIA – BY THE NUMBERS (CONTINUED)

Measure	2015	Latest	Change since 2015	U.S., latest	U.S. change since 2015	Rank	Source
INTERNATIONAL STUDENTS (CONTINUED)							
International share of undergraduate enrollment	3.9%	**3.9%**[2024]	0.0 pp	3.0%	−0.3 pp ▼	#9	[4]
International share of graduate enrollment	16%	**18%**[2024]	+2.1 pp ▲	17%	+2.8 pp ▲	#17	[4]
International share of all bachelor's	3.6%	**4.8%**[2025]	+1.2 pp ▲	4.1%	+0.1 pp ▲	#10	[4]
International share of STEM bachelor's	6.1%	**8.3%**[2025]	+2.2 pp ▲	6.6%	+0.8 pp ▲	#6	[4]
International share of all master's	16%	**24%**[2025]	+7.2 pp ▲	22%	+6.2 pp ▲	#12	[4]
International share of STEM master's	47%	**54%**[2025]	+6.9 pp ▲	52%	+9.1 pp ▲	#14	[4]
International share of computer & information sciences master's	63%	**62%**[2025]	−1.1 pp ▼	65%	+10.1 pp ▲	#27	[4]
International share of engineering & eng. technology master's	49%	**56%**[2025]	+7.6 pp ▲	50%	+0.9 pp ▲	#9	[4]
International share of mathematics & statistics master's	33%	**56%**[2025]	+23.5 pp ▲	55%	+6.1 pp ▲	#13	[4]
International share of biological & biomedical sciences master's	23%	**19%**[2025]	−4.2 pp ▼	23%	+6.2 pp ▲	#22	[4]
International share of physical sciences master's	25%	**35%**[2025]	+9.9 pp ▲	33%	+3.6 pp ▲	#18	[4]
International share of all research doctorates	31%	**33%**[2025]	+1.8 pp ▲	25%	−1.0 pp ▼	#9	[4]
International share of STEM research doctorates	43%	**45%**[2025]	+1.3 pp ▲	43%	−0.5 pp ▼	#18	[4]
International share of computer & information sciences research doctorates	49%	**65%**[2025]	+15.9 pp ▲	56%	+2.0 pp ▲	#7	[4]
International share of engineering & eng. technology research doctorates	55%	**54%**[2025]	−0.4 pp ▼	55%	−1.0 pp ▼	#23	[4]
International share of mathematics & statistics research doctorates	59%	**50%**[2025]	−9.1 pp ▼	53%	+2.9 pp ▲	#18	[4]
International share of biological & biomedical sciences research doctorates	25%	**21%**[2025]	−3.7 pp ▼	23%	−4.3 pp ▼	#29	[4]
International share of physical sciences research doctorates	39%	**42%**[2025]	+3.1 pp ▲	38%	−1.1 pp ▼	#17	[4]
Active F-1/M-1 students, all levels	52.4K	**48.6K**[2026]	−7.3% ▼	1.19M	−0.1% ▼	#7	[20]
Share of U.S. active F-1/M-1 students	4.4%	**4.1%**[2026]	−0.3 pp ▼	–	–	#7	[20]
Share of F-1/M-1 students from India	15%	**22%**[2026]	+6.9 pp ▲	27%	+11.4 pp ▲	#24	[20]
Share of F-1/M-1 students from China	39%	**27%**[2026]	−11.7 pp ▼	18%	−11.7 pp ▼	#5	[20]
Share from the two largest countries of origin	54%	**49%**[2026]	−4.8 pp ▼	45%	−0.3 pp ▼	#15	[20]

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PENNSYLVANIA – BY THE NUMBERS (CONTINUED)

Measure	2015	Latest	Change since 2015	U.S., latest	U.S. change since 2015	Rank	Source
INTERNATIONAL STUDENTS (CONTINUED)							
Countries of citizenship represented	185	**186**[2026]	+0.5% ▲	219	−2.7% ▼	#6	[20]
Active STEM F-1/M-1 students	24.5K	**31.1K**[2026]	+27% ▲	749K	+61% ▲	#7	[20]
Active STEM master's students	10.8K	**14.6K**[2026]	+36% ▲	393K	+90% ▲	#7	[20]
Active STEM doctoral students	4.99K	**7.08K**[2026]	+42% ▲	158K	+61% ▲	#6	[20]
Share of STEM master's students from India	43%	**46%**[2026]	+3.2 pp ▲	58%	−1.8 pp ▼	#37	[20]
U.S. students studying abroad for credit, from the state's institutions	21K[2017]	**15.8K**[2024]	−25% ▼	–	–	#4	[1]
Economic contribution of international students, nominal dollars	$2.1B[2018]	**$2.1B**[2024]	+0.0%	$42.9B	+4.6% ▲	#6	[2]
Jobs supported by international students' spending	27.5K[2018]	**20.2K**[2024]	−27% ▼	356K	−22% ▼	#6	[2]
Share of U.S. student and exchange-visitor admissions, by destination	3.2%	**3.4%**[2024]	+0.3 pp ▲	–	–	#8	[16]
Share of U.S. F-1 student admissions, by destination	3.2%	**3.4%**[2024]	+0.2 pp ▲	–	–	#8	[16]
Share of U.S. J-1 exchange-visitor admissions, by destination	3.2%	**3.9%**[2024]	+0.8 pp ▲	–	–	#7	[16]
J-1 exchange visitors starting a program in the state	12.6K	**13.1K**[2025]	+4.4% ▲	275K	−14% ▼	#4	[18]
GRADUATE EDUCATION AND RESEARCH							
Graduate students in science & engineering	26.4K	**28.4K**[2016]	+7.8% ▲	620K	+0.4% ▲	#6	[9]
S&E graduate students on temporary visas	10.7K	**12.5K**[2016]	+17% ▲	241K	+2.3% ▲	#5	[9]
Temporary-visa share of S&E graduate students	41%	**44%**[2016]	+3.6 pp ▲	39%	+0.7 pp ▲	#10	[9]
Temporary-visa share of engineering graduate students	60%	**60%**[2016]	−0.3 pp ▼	57%	0.0 pp	#12	[9]
S&E postdoctoral appointees	1.85K	**1.99K**[2016]	+8.0% ▲	45.7K	+1.0% ▲	#5	[9]
Temporary-visa share of S&E postdocs	59%	**58%**[2016]	−0.5 pp ▼	56%	−0.9 pp ▼	#18	[9]
University R&D spending, nominal dollars	$3.36B	**$6.45B**[2024]	+92% ▲	$118B	+71% ▲	#5	[6]
Share of U.S. university R&D	4.9%	**5.5%**[2024]	+0.6 pp ▲	–	–	#5	[6]
Federally funded share of university R&D	65%	**56%**[2024]	−8.9 pp ▼	55%	−0.2 pp ▼	#17	[6]
Business-funded share of university R&D	5.9%	**5.8%**[2024]	−0.1 pp ▼	5.4%	−0.4 pp ▼	#12	[6]
State- and locally funded share of university R&D	2.4%	**2.0%**[2024]	−0.3 pp ▼	5.2%	−0.4 pp ▼	#44	[6]
University R&D in mathematics and computer science, nominal dollars	$220M	**$476M**[2024]	2.16× ▲	$5.12B	+97% ▲	#2	[6]

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PENNSYLVANIA – BY THE NUMBERS (CONTINUED)

Measure	2015	Latest	Change since 2015	U.S., latest	U.S. change since 2015	Rank	Source
GRADUATE EDUCATION AND RESEARCH (CONTINUED)							
University R&D in engineering, nominal dollars	$543M	**$952M**[2024]	+75% ▲	$19.3B	+74% ▲	#7	[6]
University R&D personnel, headcount	51.8K	**51.4K**[2024]	−0.9% ▼	1.09M	+20% ▲	#4	[6]
Federal R&D obligations to the state, nominal dollars	$3.71B	**$5.15B**[2024]	+39% ▲	$193B	+53% ▲	#12	[8]
Federal R&D obligations to universities in the state, nominal dollars	$1.66B	**$2.56B**[2024]	+54% ▲	$44.4B	+64% ▲	#5	[8]
Defense R&D obligations to the state, nominal dollars	$1.24B	**$1.58B**[2024]	+27% ▲	$97.2B	+60% ▲	#13	[8]
Health (HHS/NIH) R&D obligations to the state, nominal dollars	$1.52B	**$2.26B**[2024]	+48% ▲	$48.5B	+62% ▲	#5	[8]
NSF, NIH, NASA and DOE Science grants performed in the state, nominal dollars	$1.82B	**$2.57B**[2025]	+42% ▲	$45B	+47% ▲	#4	[19]
Federal R&D contracts performed in the state, nominal dollars	$630M	**$1.6B**[2026]	+970.3 ▲	$59.3B	+22,105.9 ▲	#9	[19]
COLLEGE COSTS AND OUTCOMES							
Median earnings 10 years after entering college	$45K[2020]	**$71.8K**[2022]	+59% ▲	$47.5K	+11% ▲	#1	[15]
Median earnings 1 year after completing	–	**$50.4K**[2025]	–	$47.5K	–	#16	[15]
Median debt of completers, nominal dollars	$26K	**$23.3K**[2024]	−11% ▼	$17.6K	−4.6% ▼	#2	[15]
STEM bachelor's graduates: median earnings 1 year out	–	**$68.2K**[2025]	–	$68.2K	–	#22	[15]
Non-STEM bachelor's graduates: median earnings 1 year out	–	**$49.4K**[2025]	–	$51.9K	–	#31	[15]
Bachelor's graduates: median earnings 5 years out	–	**$69.5K**[2025]	–	$68.1K	–	#15	[15]
STEM master's graduates: median earnings 1 year out	–	**$103K**[2025]	–	$91.6K	–	#5	[15]
Six-year bachelor's graduation rate, 4-year institutions	67%	**68%**[2024]	+0.9 pp ▲	65%	+5.3 pp ▲	#15	[4]
Completion within 150% of normal time, 2-year institutions	37%	**42%**[2024]	+5.4 pp ▲	41%	+12.1 pp ▲	#22	[4]
In-state tuition & fees, public 4-year	$17.8K	**$17K**[2023]	−4.3% ▼	$9.71K	−14% ▼	#3	[4]
State & local appropriations per FTE, public institutions	$5.97K	**$7.14K**[2024]	+20% ▲	$11.4K	+22% ▲	#46	[4]
Net tuition share of tuition plus state and local appropriations, public institutions	76%	**72%**[2024]	−4.0 pp ▼	43%	−5.9 pp ▼	#3	[4]
Average full-time faculty salary (9/10-month)	$119K[2016]	**$109K**[2024]	−8.2% ▼	$106K	−4.6% ▼	#14	[4]
WHERE STUDENTS AND GRADUATES GO							

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PENNSYLVANIA – BY THE NUMBERS (CONTINUED)

Measure	2015	Latest	Change since 2015	U.S., latest	U.S. change since 2015	Rank	Source
WHERE STUDENTS AND GRADUATES GO (CONTINUED)							
International graduates on OPT working in the state where they studied	29%	**26%**[2016]	−2.6 pp ▼	44%	−1.8 pp ▼	#38	[21]
OPT retention, bachelor's graduates	37%	**30%**[2016]	−6.9 pp ▼	58%	+0.3 pp ▲	#44	[21]
OPT retention, master's graduates	25%	**24%**[2016]	−1.4 pp ▼	39%	−2.5 pp ▼	#35	[21]
OPT retention, STEM graduates	25%	**22%**[2016]	−2.3 pp ▼	37%	−1.9 pp ▼	#35	[21]
OPT workers employed in the state per OPT graduate of its schools	0.58	**0.58**[2016]	0.0	1.00	0.0	#37	[21]
STEM OPT extensions approved, by state of the school	1.56K	**1.95K**[2016]	+25% ▲	41.2K	+51% ▲	#5	[21]
Bachelor's graduates working in the state 1 year after graduating (PSEO)	55%[2016]	**55%**[2019]	+0.2 pp ▲	72%	−0.1 pp ▼	#27	[11]
Bachelor's graduates working in the state 5 years after graduating (PSEO)	–	**51%**[2016]	–	65%	–	#26	[11]
Bachelor's graduates working in the state 10 years after graduating (PSEO)	–	**51%**[2010]	–	64%	–	#26	[11]
STEM bachelor's graduates working in the state 1 year out (PSEO)	52%[2016]	**55%**[2019]	+2.7 pp ▲	67%	+1.5 pp ▲	#23	[11]
Master's graduates working in the state 1 year out (PSEO)	–	**48%**[2016]	–	66%	–	#31	[11]
Doctoral graduates working in the state 1 year out (PSEO)	–	**40%**[2016]	–	49%	–	#27	[11]
Bachelor's graduates' median earnings 1 year out (PSEO)	$54.8K[2016]	**$55K**[2019]	+0.2% ▲	$48.1K	+2.3% ▲	#2	[11]
Bachelor's graduates' median earnings 5 years out (PSEO)	–	**$79.5K**[2016]	–	$67.3K	–	#2	[11]
STEM bachelor's graduates' median earnings 1 year out (PSEO)	$65.4K[2016]	**$64.9K**[2019]	−0.8% ▼	$59.3K	+0.1% ▲	#3	[11]
Net domestic migration of bachelor's-plus adults, per 1,000 residents	−4.9	**−2.6**[2025]	+2.3 ▲	–	–	#34	[12]
Net domestic migration of graduate-degree holders	−10.9K	**−10.3K**[2025]	+586.0 ▲	–	–	#50	[12]
Bachelor's-plus adults who moved in from another state	2.4%	**2.2%**[2025]	−0.2 pp ▼	–	–	#38	[12]
Bachelor's-plus adults who moved out to another state	2.9%	**2.4%**[2025]	−0.5 pp ▼	–	–	#34	[12]
Bachelor's-plus adults who arrived from abroad in the past year	16.8K	**18K**[2025]	+7.1% ▲	–	–	#11	[12]
First-time students who are in-state residents	71%[2016]	**68%**[2024]	−2.4 pp ▼	–	–	#31	[4]
Net student migration (per 100 resident first-time students)	+11[2016]	**+8**[2024]	−2.9 ▼	–	–	#21	[4]

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PENNSYLVANIA – BY THE NUMBERS (CONTINUED)

Measure	2015	Latest	Change since 2015	U.S., latest	U.S. change since 2015	Rank	Source
WHERE STUDENTS AND GRADUATES GO (CONTINUED)							
Recent high school graduates who go to college out of state	20%[2016]	**27%[2022]**	+7.6 pp ▲	22%	+2.5 pp ▲	#19	[3]
WORK VISAS AND GREEN CARDS							
Certified H-1B positions, by worksite	63.8K	**20K[2025]**	−69% ▼	914K	−24% ▼	#12	[17]
Share of U.S. certified H-1B positions	5.3%	**2.2%[2025]**	−3.1 pp ▼	–	–	#12	[17]
STEM share of certified H-1B positions	84%	**79%[2025]**	−4.5 pp ▼	76%	−7.8 pp ▼	#11	[17]
Median offered H-1B wage, nominal dollars, nominal dollars	$69.5K	**$103K[2025]**	+49% ▲	$119K	+66% ▲	#27	[17]
H-1B positions at H-1B-dependent employers	31%	**24%[2025]**	−6.9 pp ▼	22%	−25.7 pp ▼	#17	[17]
Green-card labor certifications	1.8K	**3.18K[2025]**	+77% ▲	138K	+75% ▲	#13	[17]
Share of PERM certifications for citizens of India	64%	**45%[2024]**	−18.7 pp ▼	47%	−10.6 pp ▼	#19	[17]
PERM certifications for workers on F-1 student visas	2.4%	**9.9%[2024]**	+7.6 pp ▲	8.7%	+4.9 pp ▲	#24	[17]
PERM workers with a master's or doctorate	53%	**44%[2024]**	−8.5 pp ▼	47%	−4.9 pp ▼	#20	[17]
STEM share of PERM certifications	71%	**51%[2025]**	−20.3 pp ▼	49%	−25.8 pp ▼	#14	[17]
Median offered PERM wage, nominal dollars, nominal dollars	$90.5K	**$94.7K[2025]**	+4.7% ▲	$100K	+9.9% ▲	#20	[17]
New lawful permanent residents (green cards), by state of residence	25K	**34.9K[2024]**	+40% ▲	1.36M	+30% ▲	#10	[16]
Share of U.S. new green-card holders	2.4%	**2.6%[2024]**	+0.2 pp ▲	–	–	#10	[16]
People naturalized as U.S. citizens, by state of residence	16.6K	**22.2K[2024]**	+34% ▲	819K	+12% ▲	#11	[16]
Share of U.S. naturalizations	2.3%	**2.7%[2024]**	+0.5 pp ▲	–	–	#11	[16]
Share of U.S. temporary-worker admissions, by destination	1.5%	**1.3%[2024]**	−0.2 pp ▼	–	–	#13	[16]
Share of U.S. H-1B worker admissions, by destination	2.3%	**1.9%[2024]**	−0.4 pp ▼	–	–	#12	[16]
Share of U.S. L-1 intracompany-transferee admissions, by destination	2.0%	**1.8%[2024]**	−0.2 pp ▼	–	–	#12	[16]
Share of U.S. O-1 extraordinary-ability worker admissions, by destination	1.0%	**1.1%[2024]**	+0.1 pp ▲	–	–	#12	[16]
New H-1B approvals (initial), by employer state	3.38K	**3.6K[2026]**	+6.7% ▲	127K	+14% ▲	#9	[13]
Continuing H-1B approvals, by employer state	5.76K	**8.55K[2026]**	+48% ▲	268K	+52% ▲	#8	[13]
Share of U.S. new H-1B approvals	3.0%	**2.8%[2026]**	−0.2 pp ▼	–	–	#9	[13]

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PENNSYLVANIA – BY THE NUMBERS (CONTINUED)

Measure	2015	Latest	Change since 2015	U.S., latest	U.S. change since 2015	Rank	Source
WORK VISAS AND GREEN CARDS (CONTINUED)							
Denial rate, new H-1B petitions	10%	**2.7%**[2026]	−7.8 pp ▼	3.4%	−3.0 pp ▼	#41	[13]
New H-1B approvals at colleges and schools (NAICS 61)	17%	**13%**[2026]	−4.3 pp ▼	8.6%	−1.7 pp ▼	#30	[13]
New H-1B approvals in professional, scientific & technical services	37%	**50%**[2026]	+13.4 pp ▲	44%	−12.5 pp ▼	#9	[13]
Five largest employers' share of new H-1B approvals	32%	**26%**[2026]	−6.7 pp ▼	9.1%	−6.7 pp ▼	#35	[13]
STEM WORKFORCE AND INNOVATION							
Workers in STEM occupations	344K	**521K**[2024]	+51% ▲	13.7M	+51% ▲	#5	[12]
STEM workers as a share of all workers	5.7%	**8.0%**[2024]	+2.4 pp ▲	8.1%	+2.1 pp ▲	#20	[12]
Foreign-born share of STEM workers	14%	**16%**[2024]	+2.2 pp ▲	24%	+1.6 pp ▲	#21	[12]
Foreign-born share of STEM workers with a graduate degree	29%	**29%**[2024]	−0.2 pp ▼	41%	+2.2 pp ▲	#23	[12]
Foreign-born share of STEM workers with a doctorate	46%	**29%**[2024]	−17.6 pp ▼	47%	+0.9 pp ▲	#19	[12]
Noncitizen share of STEM workers	7.2%	**7.5%**[2024]	+0.3 pp ▲	11%	+0.7 pp ▲	#22	[12]
STEM workers born abroad who arrived in the last 10 years	5.0%	**4.6%**[2024]	−0.4 pp ▼	7.5%	+0.3 pp ▲	#30	[12]
Share of foreign-born STEM workers born in India	35%	**35%**[2024]	−0.1 pp ▼	28%	+0.7 pp ▲	#7	[12]
Foreign-born share of residents	6.5%	**8.4%**[2025]	+1.9 pp ▲	15%	+1.3 pp ▲	#26	[12]
Foreign-born share of graduate-degree holders, 25+	13%	**15%**[2025]	+2.6 pp ▲	21%	+2.8 pp ▲	#20	[12]
STEM jobs, employer survey	310K	**377K**[2025]	+22% ▲	10.4M	+26% ▲	#7	[10]
STEM jobs concentration vs the U.S., location quotient	0.91	**0.93**[2025]	0.0	1.00	0.0	#22	[10]
Mean annual STEM wage, nominal, nominal dollars	$79.8K	**$109K**[2025]	+37% ▲	$120K	+38% ▲	#22	[10]
Median software developer wage, nominal, nominal dollars	$96.5K[2019]	**$127K**[2025]	+31% ▲	$136K	+26% ▲	#26	[10]
Software developer jobs	32.5K	**55.9K**[2025]	+72% ▲	1.69M	+48% ▲	#9	[10]
Mean annual wage, all occupations, nominal, nominal dollars	$46.5K	**$65.7K**[2025]	+41% ▲	$69.8K	+44% ▲	#25	[10]
Patents granted to assignees in the state	2.77K	**3.2K**[2025]	+16% ▲	135K	−0.6% ▼	#12	[22]
Share of U.S.-assignee patents	2.0%	**2.4%**[2025]	+0.3 pp ▲	–	–	#12	[22]
Share of the state's patents in digital technologies	16%	**28%**[2025]	+12.5 pp ▲	43%	+1.6 pp ▲	#17	[22]
Share of the state's patents in life sciences	27%	**28%**[2025]	+1.6 pp ▲	17%	+0.4 pp ▲	#10	[22]

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PENNSYLVANIA – BY THE NUMBERS (CONTINUED)

Measure	2015	Latest	Change since 2015	U.S., latest	U.S. change since 2015	Rank	Source
STEM WORKFORCE AND INNOVATION (CONTINUED)							
Share of the state's patents in semiconductors	1.7%	**0.8%**^{2025}	−0.9 pp ▼	2.9%	−1.7 pp ▼	#23	[22]
Women's share of inventors on the state's patents	10%	**12%**^{2025}	+1.2 pp ▲	14%	+3.4 pp ▲	#25	[22]

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Sources

[1] Institute of International Education. *Open Doors Report on International Educational Exchange: State Fact Sheets*. Editions 2020–2026. New York: Institute of International Education. https://opendoorsdata.org/. Accessed November 2026.

[2] NAFSA: Association of International Educators. *NAFSA International Student Economic Value Tool*. Academic years 2018–19 to 2024–25. Washington, DC: NAFSA. https://www.nafsa.org/isev. Accessed November 2026.

[3] National Center for Education Statistics. *Digest of Education Statistics*. Editions 1995–2025. Washington, DC: U.S. Department of Education, Institute of Education Sciences. https://nces.ed.gov/programs/digest/. Accessed November 2026.

[4] National Center for Education Statistics. *Integrated Postsecondary Education Data System (IPEDS)*. Survey files, 1980–2025. Washington, DC: U.S. Department of Education, Institute of Education Sciences. https://nces.ed.gov/ipeds/. Accessed November 2026.

[5] National Center for Education Statistics. *National Assessment of Educational Progress (NAEP)*. NAEP Data Service, 1990–2024. Washington, DC: U.S. Department of Education, Institute of Education Sciences. https://www.nationsreportcard.gov/. Accessed November 2026.

[6] National Center for Science and Engineering Statistics. *Higher Education Research and Development Survey*. Data tables, FY2003–2024. Alexandria, VA: National Science Foundation. https://ncses.nsf.gov/surveys/higher-education-research-development. Accessed November 2026.

[7] National Center for Science and Engineering Statistics. *Survey of Earned Doctorates*. Data tables, 1994–2025. Alexandria, VA: National Science Foundation. https://ncses.nsf.gov/surveys/earned-doctorates. Accessed November 2026.

[8] National Center for Science and Engineering Statistics. *Survey of Federal Funds for Research and Development*. Data tables, FY1985–2024. Alexandria, VA: National Science Foundation. https://ncses.nsf.gov/surveys/federal-funds-research-development. Accessed November 2026.

[9] National Center for Science and Engineering Statistics. *Survey of Graduate Students and Postdoctorates in Science and Engineering*. Data tables, 1994–2024. Alexandria, VA: National Science Foundation. https://ncses.nsf.gov/surveys/graduate-students-postdoctorates-s-e. Accessed November 2026.

[10] U.S. Bureau of Labor Statistics. *Occupational Employment and Wage Statistics*. State estimates, 1999–2025. Washington, DC: U.S. Department of Labor. https://www.bls.gov/oes/. Accessed November 2026.

[11] U.S. Census Bureau, Center for Economic Studies. *Post-Secondary Employment Outcomes (PSEO)*. Release R2025. Washington, DC: U.S. Department of Commerce. https://lehd.ces.census.gov/data/pseo_experimental.html. Accessed November 2026.

[12] U.S. Census Bureau. *American Community Survey*. 1-year estimates and Public Use Microdata Sample, 2010–2025. Washington, DC: U.S. Department of Commerce. https://data.census.gov/. Accessed November 2026.

[13] U.S. Citizenship and Immigration Services. *H-1B Employer Data Hub*. Fiscal years 2009–2026. Washington, DC: U.S. Department of Homeland Security. https://www.uscis.gov/tools/reports-and-studies/h-1b-employer-data-hub. Accessed November 2026.

[14] U.S. Department of Education, Office for Civil Rights. *Civil Rights Data Collection*. School years 1980–81 to 2023–24. Washington, DC: U.S. Department of Education. https://civilrightsdata.ed.gov/. Accessed November 2026.

[15] U.S. Department of Education. *College Scorecard*. Institution and field-of-study data, 1996–2025. Washington, DC: U.S. Department of Education. https://collegescorecard.ed.gov/data/. Accessed November 2026.

[16] U.S. Department of Homeland Security, Office of Homeland Security Statistics. *Yearbook of Immigration Statistics*. Fiscal years 1996–2024. Washington, DC: U.S. Department of Homeland Security. https://ohss.dhs.gov/topics/immigration/yearbook. Accessed November 2026.

[17] U.S. Department of Labor, Employment and Training Administration, Office of Foreign Labor Certification. *LCA and PERM Disclosure Data*. Fiscal years 2008–2026. Washington, DC: U.S. Department of Labor. https://www.dol.gov/agencies/eta/foreign-labor/performance. Accessed November 2026.

[18] U.S. Department of State, Bureau of Educational and Cultural Affairs. *BridgeUSA: Exchange Visitor Program Participants by State*. 2015–2025. Washington, DC: U.S. Department of State. https://j1visa.state.gov/. Accessed November 2026.

[19] U.S. Department of the Treasury, Bureau of the Fiscal Service. *USAspending.gov*. Award data, FY2008–2026. Washington, DC: U.S. Department of the Treasury. https://www.usaspending.gov/. Accessed November 2026.

[20] U.S. Immigration and Customs Enforcement, Student and Exchange Visitor Program. *SEVIS by the Numbers: Data Mapping Tool*. Monthly data, 2014–2026. Washington, DC: U.S. Department of Homeland Security. https://studyinthestates.dhs.gov/sevis-data-mapping-tool. Accessed November 2026.

[21] U.S. Immigration and Customs Enforcement. *Student and Exchange Visitor Information System (SEVIS) Data*. Freedom of Information Act releases, 2008–2016. Washington, DC: U.S. Department of Homeland Security. https://www.ice.gov/foia/library. Accessed November 2026.

[22] U.S. Patent and Trademark Office. *PatentsView*. Annualized patent data, grant years 1976–2025. Alexandria, VA: U.S. Patent and Trademark Office. https://patentsview.org/. Accessed November 2026.